

PV–PWR–4 power board

Rev A2 | 2026–10–05 | Schematics authored by Chinmoy Bhuyan | PV–P100/110: 4 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)

01 PC connector, command buffers, status

Interfaces PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points

File: sheets/01_pc.kicad_sch

02 Phase 1 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/02_ph1_power.kicad_sch

03 Phase 1 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/03_ph1_gda.kicad_sch

04 Phase 1 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/04_ph1_gdb.kicad_sch

05 Phase 2 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/05_ph2_power.kicad_sch

06 Phase 2 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/06_ph2_gda.kicad_sch

07 Phase 2 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/07_ph2_gdb.kicad_sch

08 Phase 3 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/08_ph3_power.kicad_sch

09 Phase 3 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/09_ph3_gda.kicad_sch

10 Phase 3 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/10_ph3_gdb.kicad_sch

11 Phase 4 power stage, bias, sensing

Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase)

File: sheets/11_ph4_power.kicad_sch

12 Phase 4 leg A gate drive (S1, S2)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/12_ph4_gda.kicad_sch

13 Phase 4 leg B gate drive (S3, S4)

gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain

File: sheets/13_ph4_gdb.kicad_sch

14 Port A power path (pv)

Varistor Y + monitor

File: sheets/14_leanA_power.kicad_sch

15 Port A sensing and interlocks

References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks

File: sheets/15_leanA_ctrl.kicad_sch

16 Port A coil drives

Contactors: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/16_leanA_drive.kicad_sch

16 Port B power path (battery)

CAUTION: not to be used for design_check() asserts every number on each build (outputs: PV–PWR–4 design check.txt), drafted @ few days from now, at build time. Domains: LIVE (BUS– = GND = AGND), GD_PxSy gate islands (functional: NSI6651, T_BIAS4), PE (AIN pads, inductor cores, heatsink NTC lugs, Y / MOV3 / MD crossings), SELV (AUX–T1 reinforced winding -> J_SELV). Coat to PD1. Shared heatsink: Tj 110 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 23.7 W (economiser) <= aux live rating 30.0 W; module aux 82 W. Full fan speed is only guaranteed while the converter is switching.

File: sheets/16_leanB_power.kicad_sch

17 Port B sensing and interlocks

References: PV–PWR–4 design check.txt, drafted @ few days from now, at build time. polarity and dV interlocks

File: sheets/17_leanB_ctrl.kicad_sch

18 Port B coil drives

Contactors + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder

File: sheets/18_leanB_drive.kicad_sch

18 Terminals, EMI filter, insulation test

DC terminals, nanocrystalline CM ring per port, X / Y capacitors, board entries (press–fit), PE links, heatsink, insulation monitor (port.lean_imd)

File: sheets/18_terminals_imd.kicad_sch

19 Live 5 V / 3.3 V, SELV connector

+24V (aux live) -> +5V and +5V_GD (LMR38020), +3V3 (TPS62130 from +5V_GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board

File: sheets/19_live5v.kicad_sch

20 AUX75 input taps, OR, film

One tap per port (+ pole, terminal side), return = BUS–: 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 2 x 3.3 uF / 1300 V film = surge absorber (AUX–HV rev C1 input block)

File: sheets/20_aux75_input.kicad_sch

21 AUX75 start–up, brown–in, OV, timer

As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 39–100 ms

File: sheets/21_aux75_start.kicad_sch

22 AUX75 PWM, SIC, clamp, AUX–T1

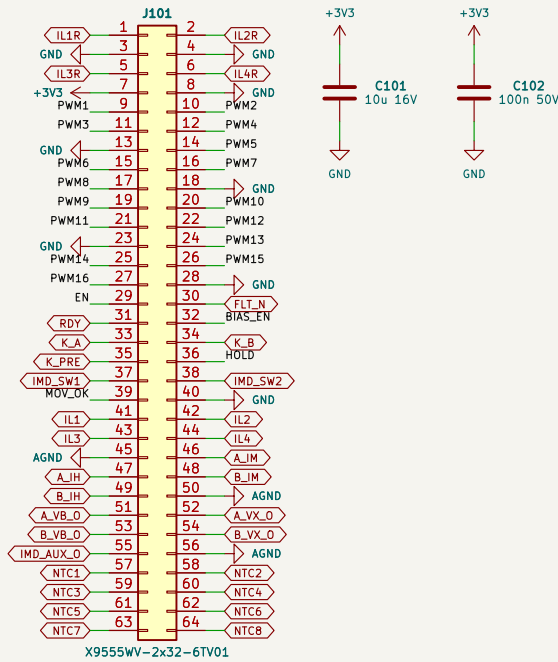
UCC28C59–Q1 64.9 kHz, IV2Q171R0D7Z, Lp 0.485 mH, Np:Nlive:Nselv 50:9:10; live 24 V regulated by the FB divider (no opto); VDD from the live 24 V through the BC817 follower; foldback detector on the live winding

File: sheets/22_aux75_pwm.kicad_sch

Not bench–validated. Values read from sim/out/*.json; checks: design_check()
Sheet 1
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

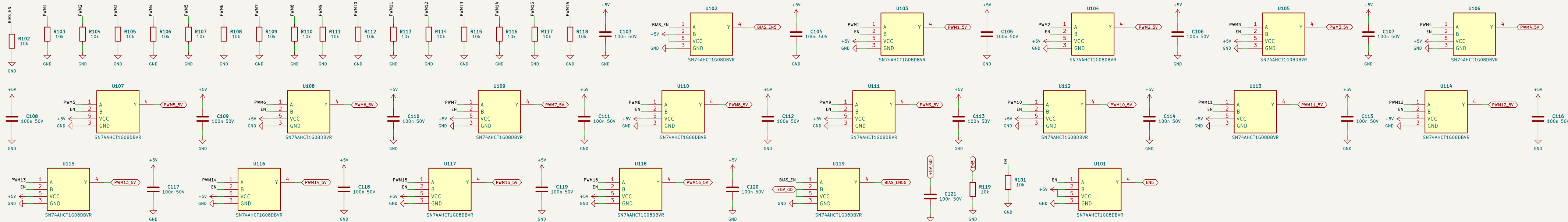
Sheets: 1
File: F
Title
Size: 6
KiCad

PC CONNECTOR (INTERFACES.PC), LIVE



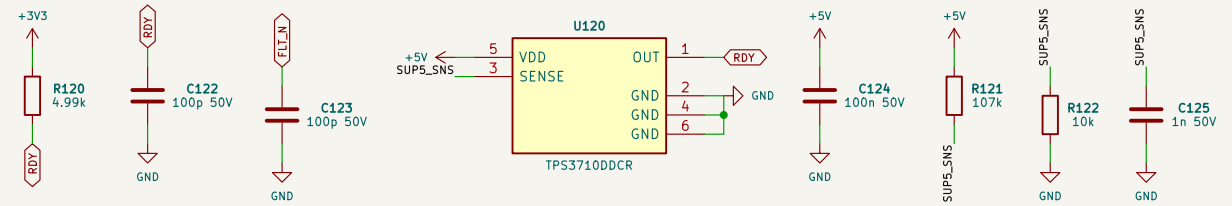
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2: the former +24V / +5V pins carry the sensor references (ILR-IL4R)). Commands are 3.3 V logic after the control board's trip latch: every command input is pulled low here (open = off).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



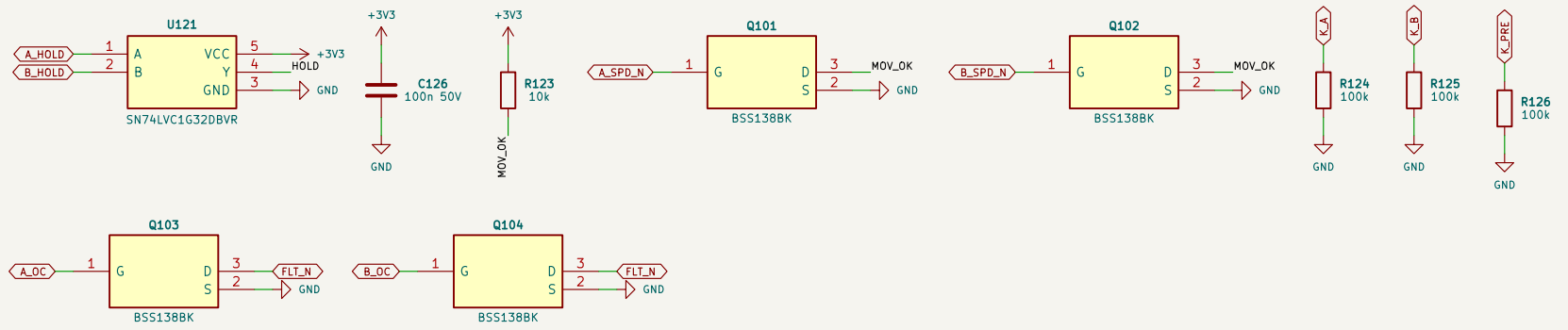
NSI6651 inputs are specified at VCC1 = 5 V only (VIN1 <= 3.3 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48–0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



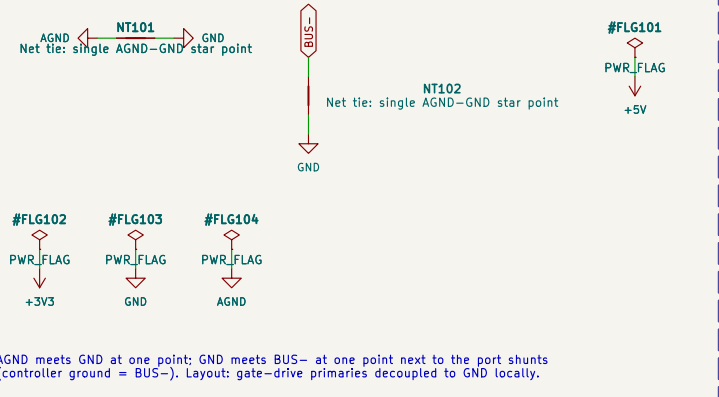
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC (I11) >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (Rd-fuse gap, port.spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



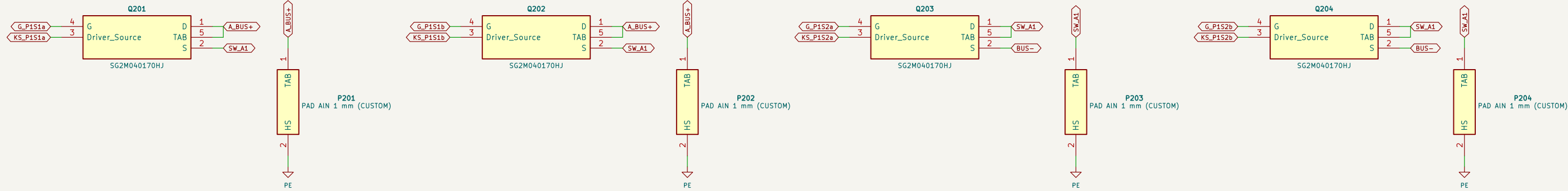
HOLD = A_HOLD OR B_HOLD (|I_port| >= 0.97–1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC (I11) >= 387–413 A, lean_shunt) pulls FLT_N low (open drain): the control board's latch trips without firmware (Rd-fuse gap, port.spec). K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



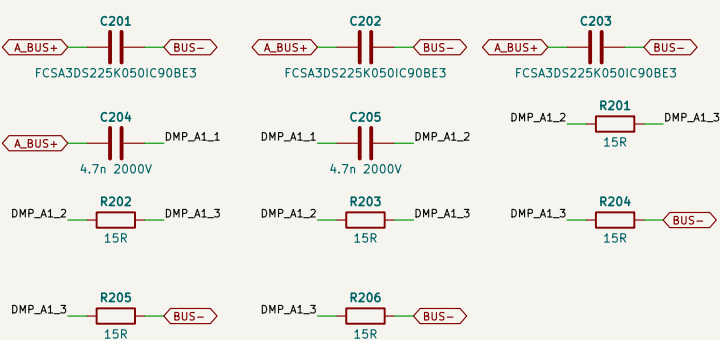
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-). Layouts gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



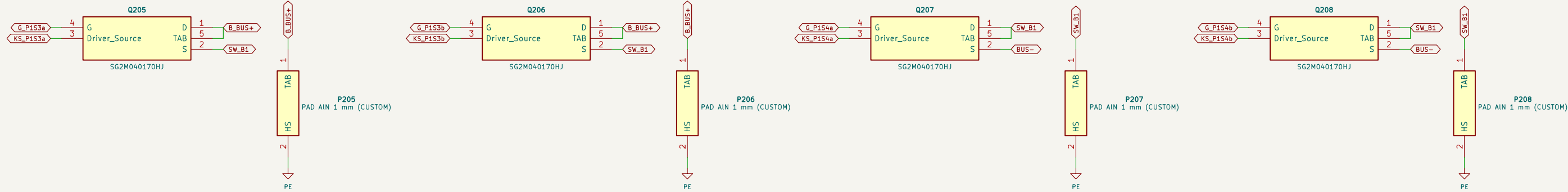
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



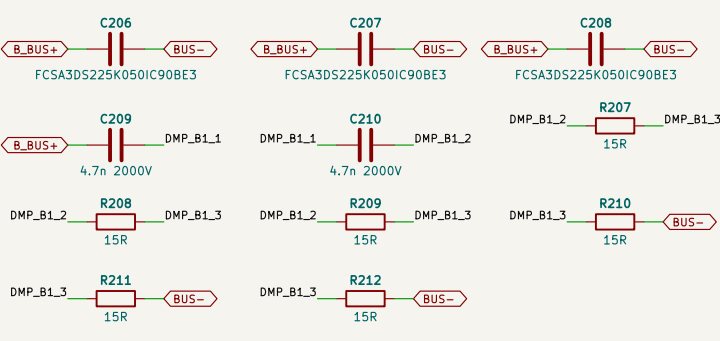
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



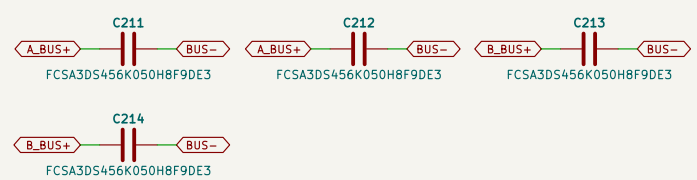
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



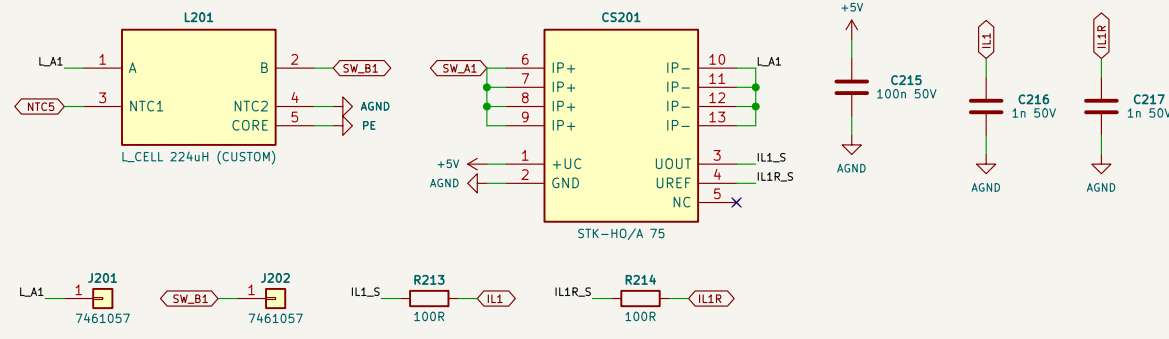
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



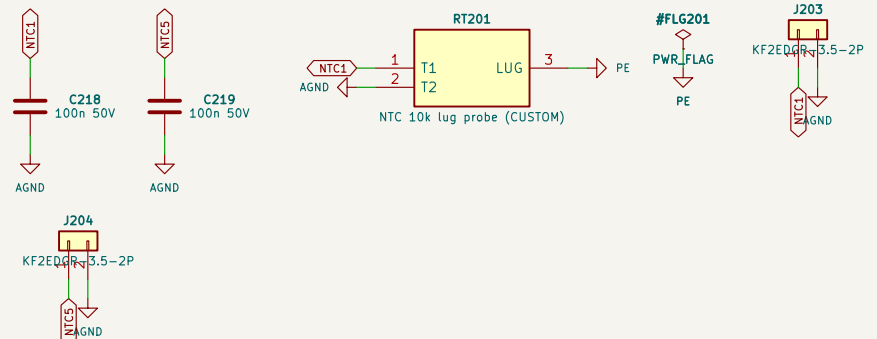
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



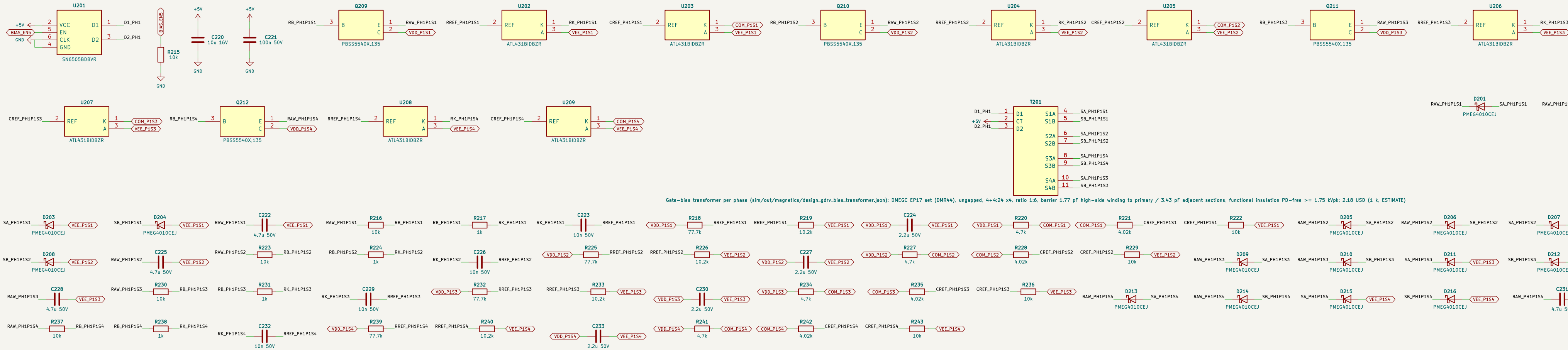
Lead A from SW_A1 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out + = SW_A -> SW_B) to stud
LA1: lead B on stud SW_B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mF at the source. IL1R = the sensor's own Uref (pin 4). 100R +/- 1 mF: PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC5 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

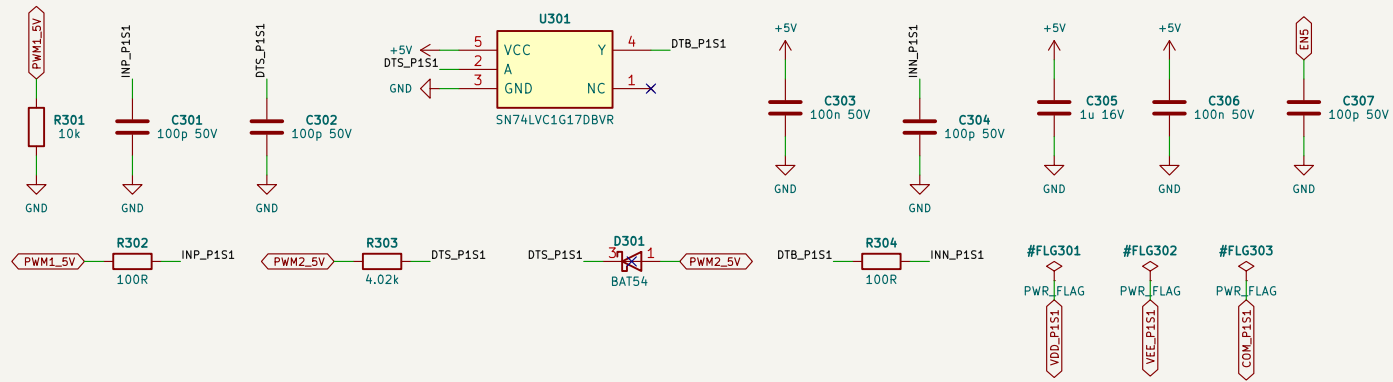
PH1: GATE BIAS (PER PHASE)



Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_bias_transformer.json): DMCC EP17 set (DMR44), ungapped, 4+4:2k v4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpk; 2.18 UOV (1 k, ESTIMATE)

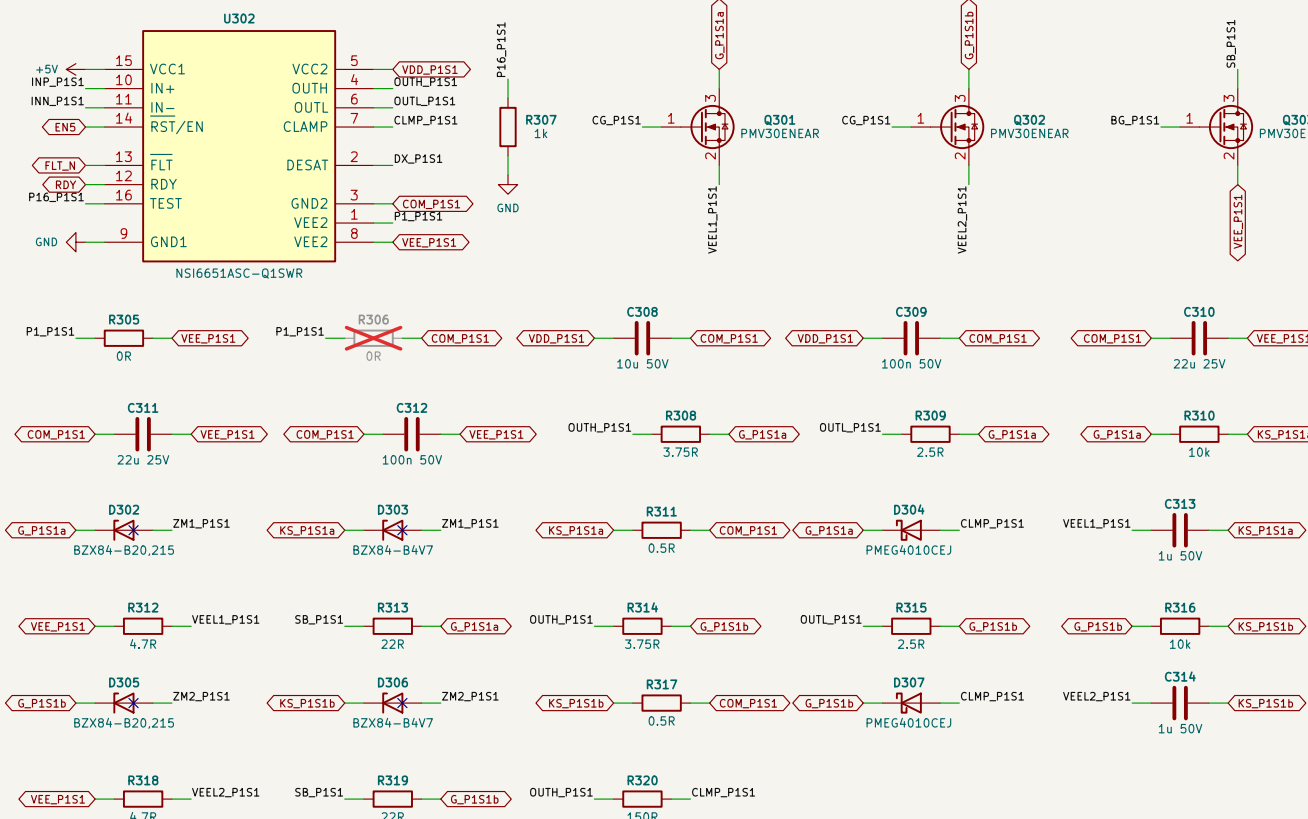
SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S1: LOGIC SIDE

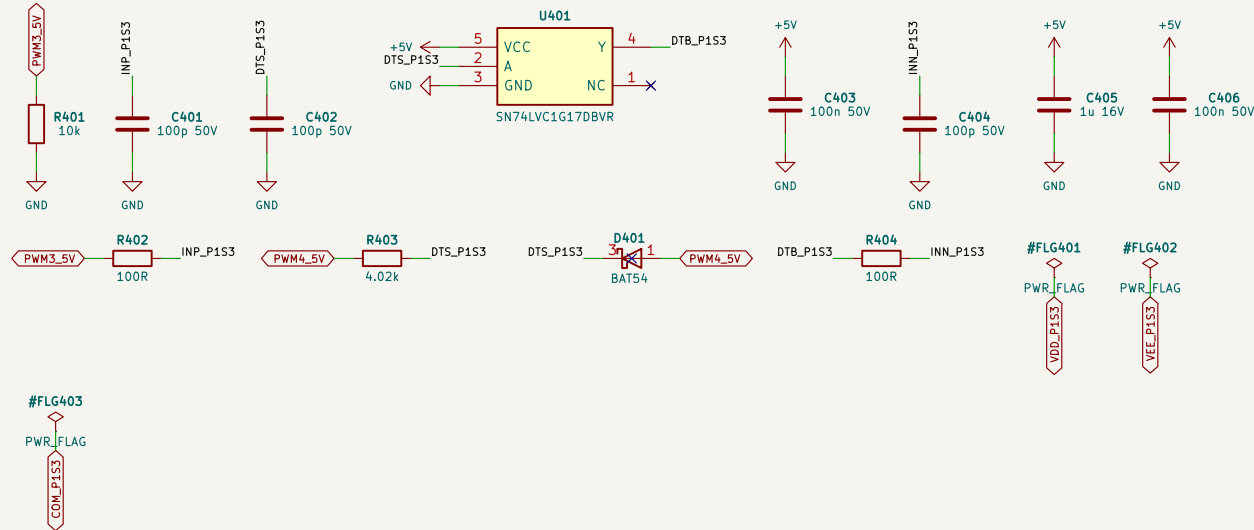


IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S1: DRIVER AND 2 GATES

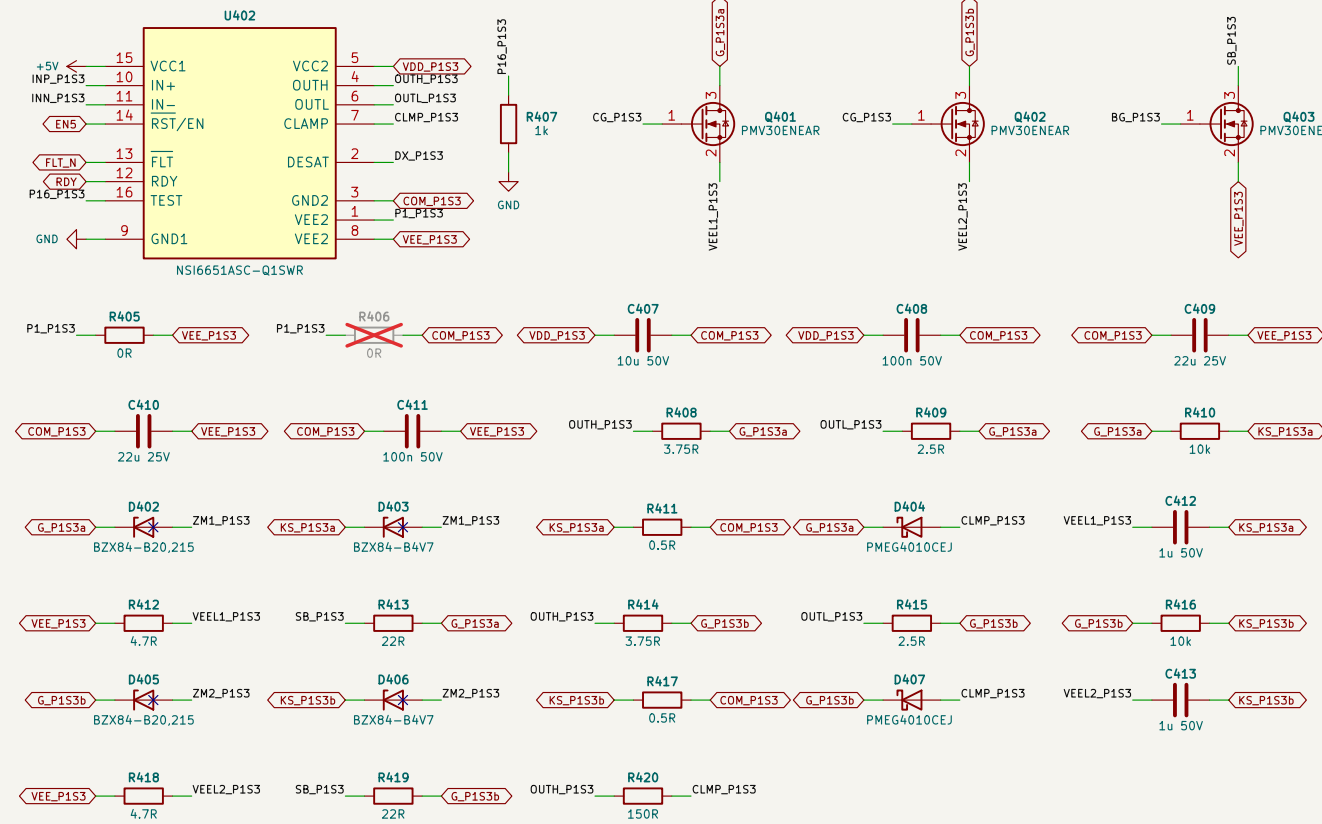


P1S3: LOGIC SIDE



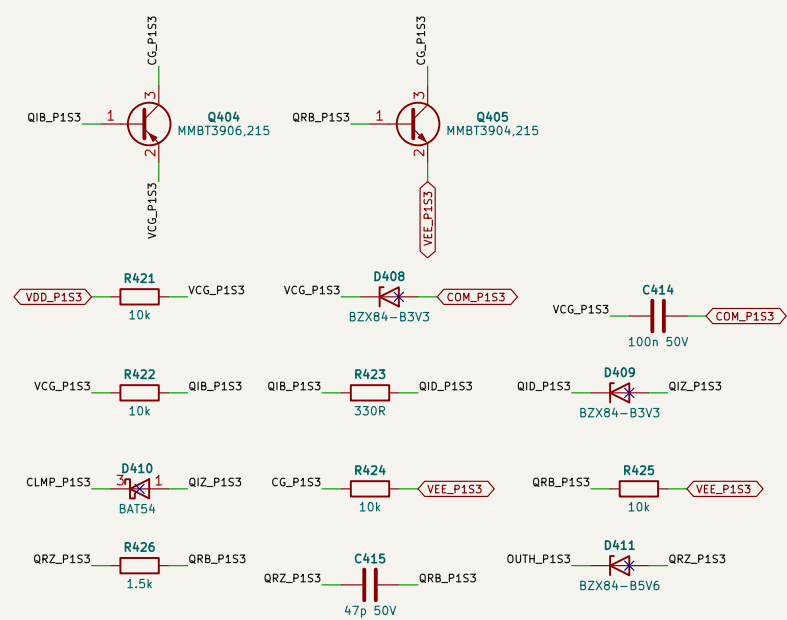
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P1S3: DRIVER AND 2 GATES



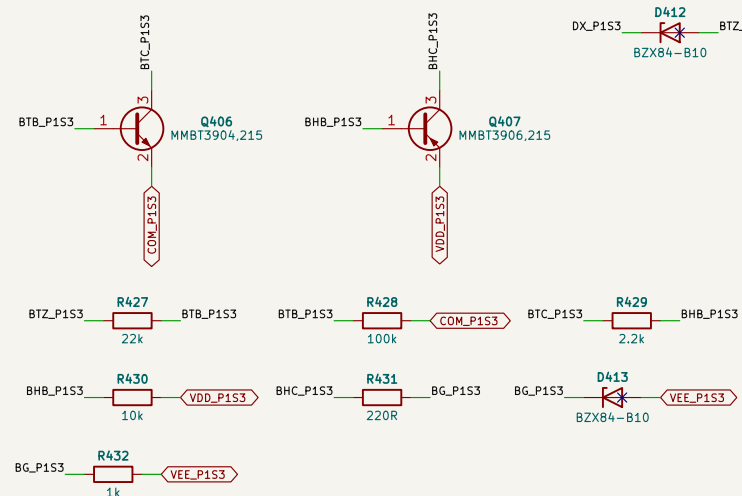
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S3: MILLER CLAMP CONTROL



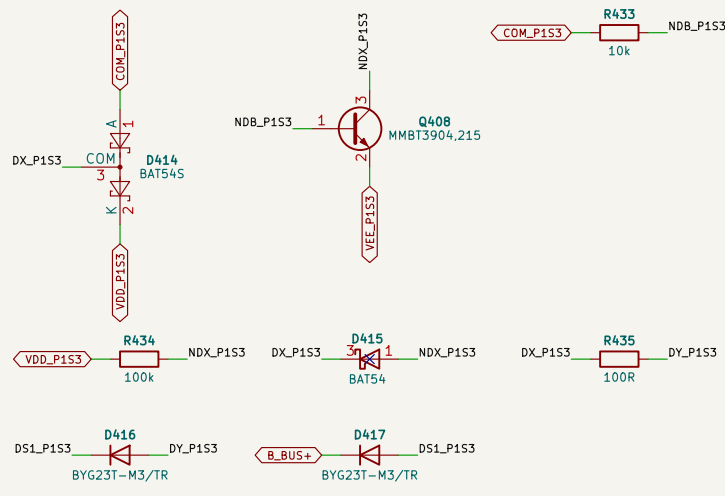
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V:
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S3: SHORT-CIRCUIT BOOSTER

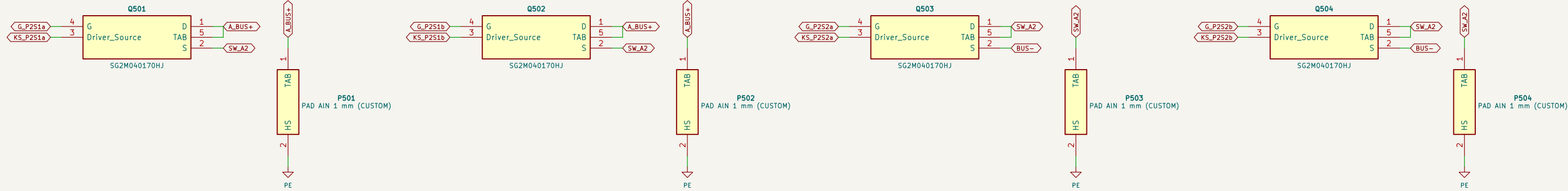


DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V:
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P1S3: DESAT

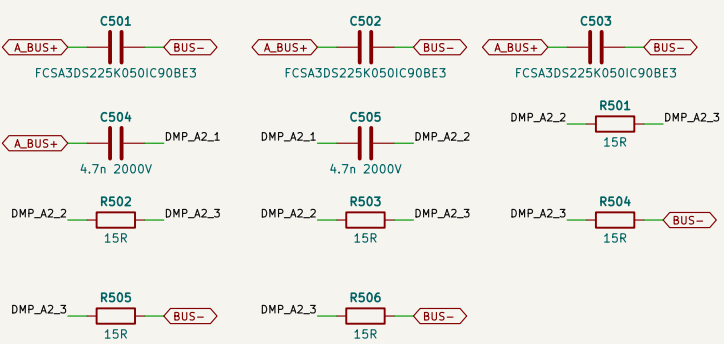


PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



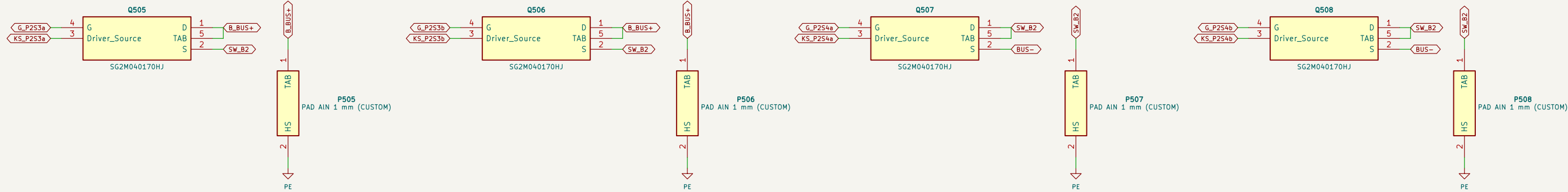
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



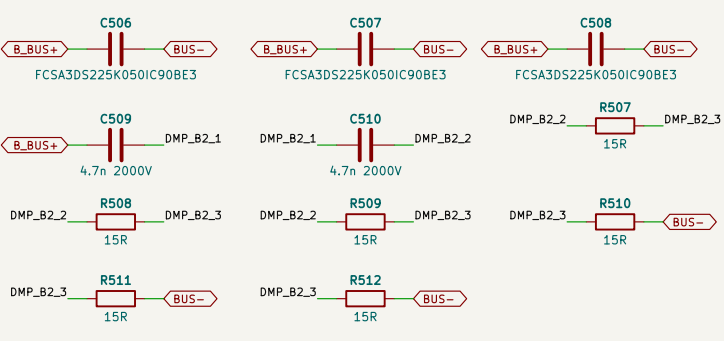
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



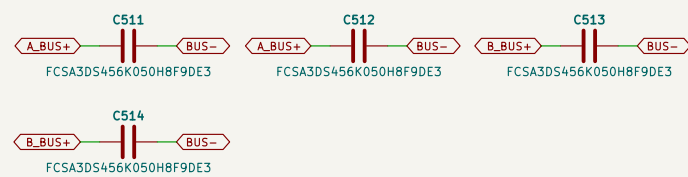
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



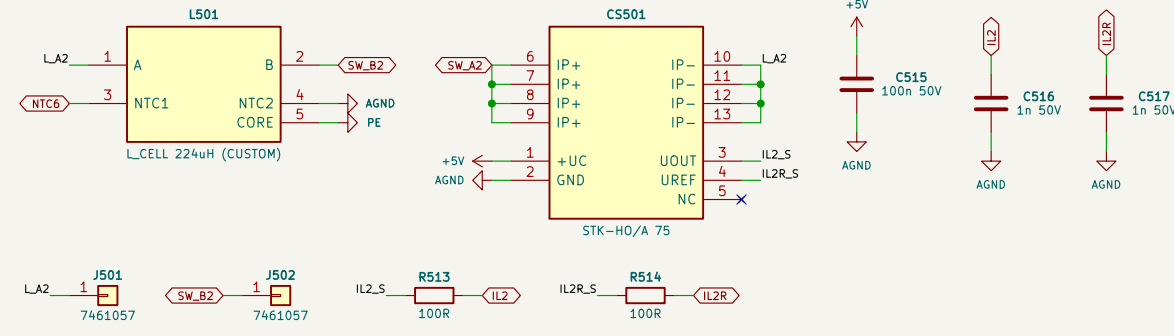
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 2 SHARE OF THE PORT BANKS



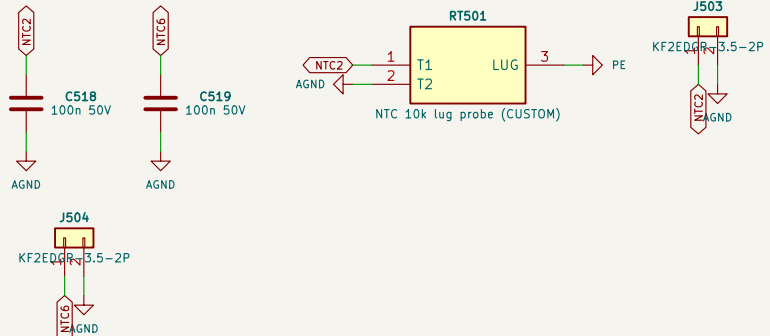
2 x FCSA3D5456 (45 uF) on A_BUS+ and 2 on B_BUS+, at this phase's legs: the banks of all phases form the port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



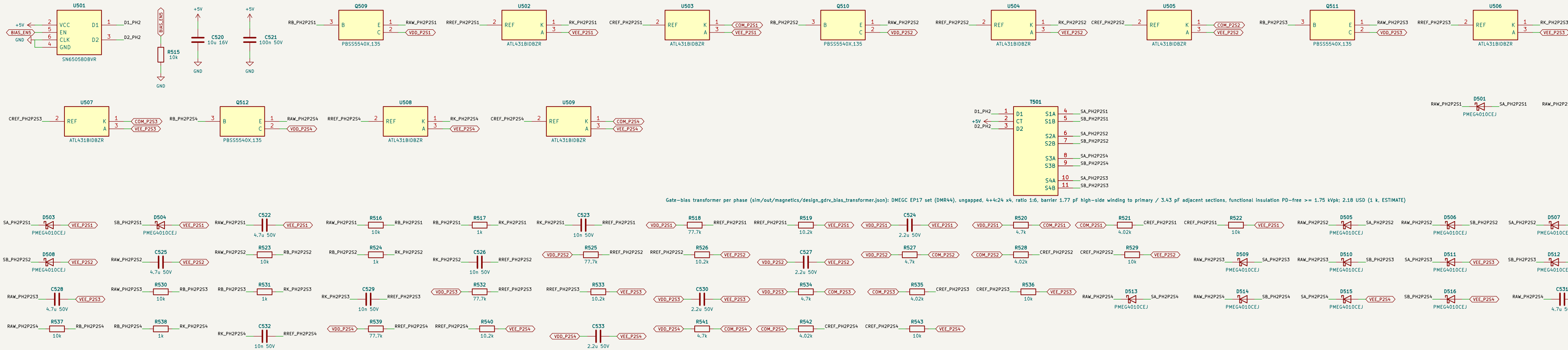
Lead A from SW_A2 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out at + = SW_A -> SW_B) to stud LA2: lead B on stud SW_B2. IL2 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A: 100R +/- 1 mF at the source. IL2R = the sensor's own Uref (pin 4). 100R +/- 1 mF: PV-CTL references its trip window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)

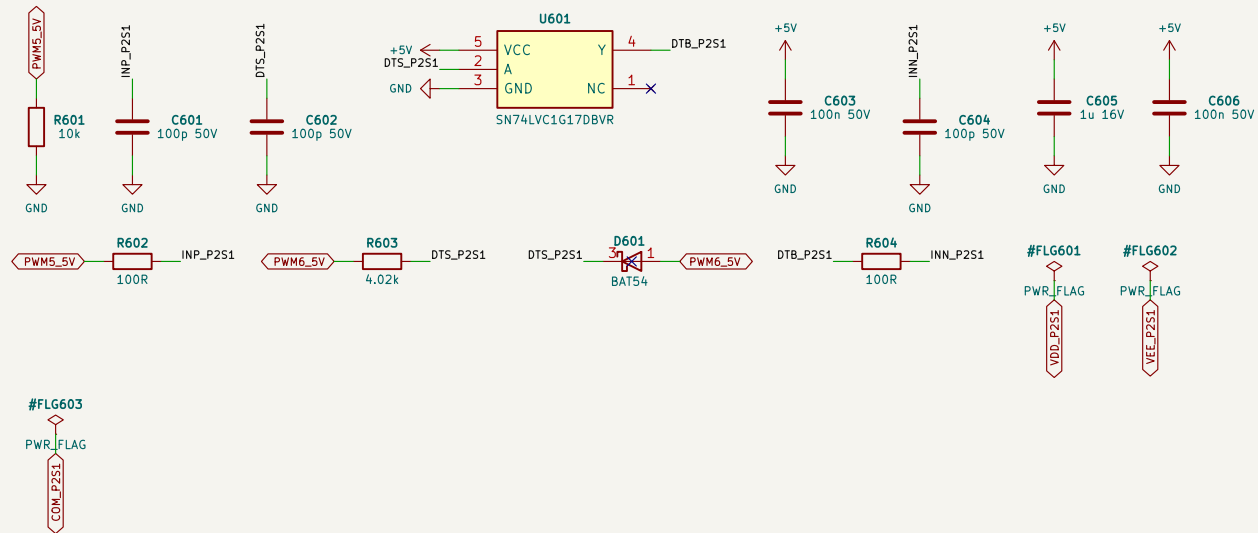


NTC2 = heatsink section of phase 2: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10). NTC6 = inductor winding NTC (functional to the winding). 100 mF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)

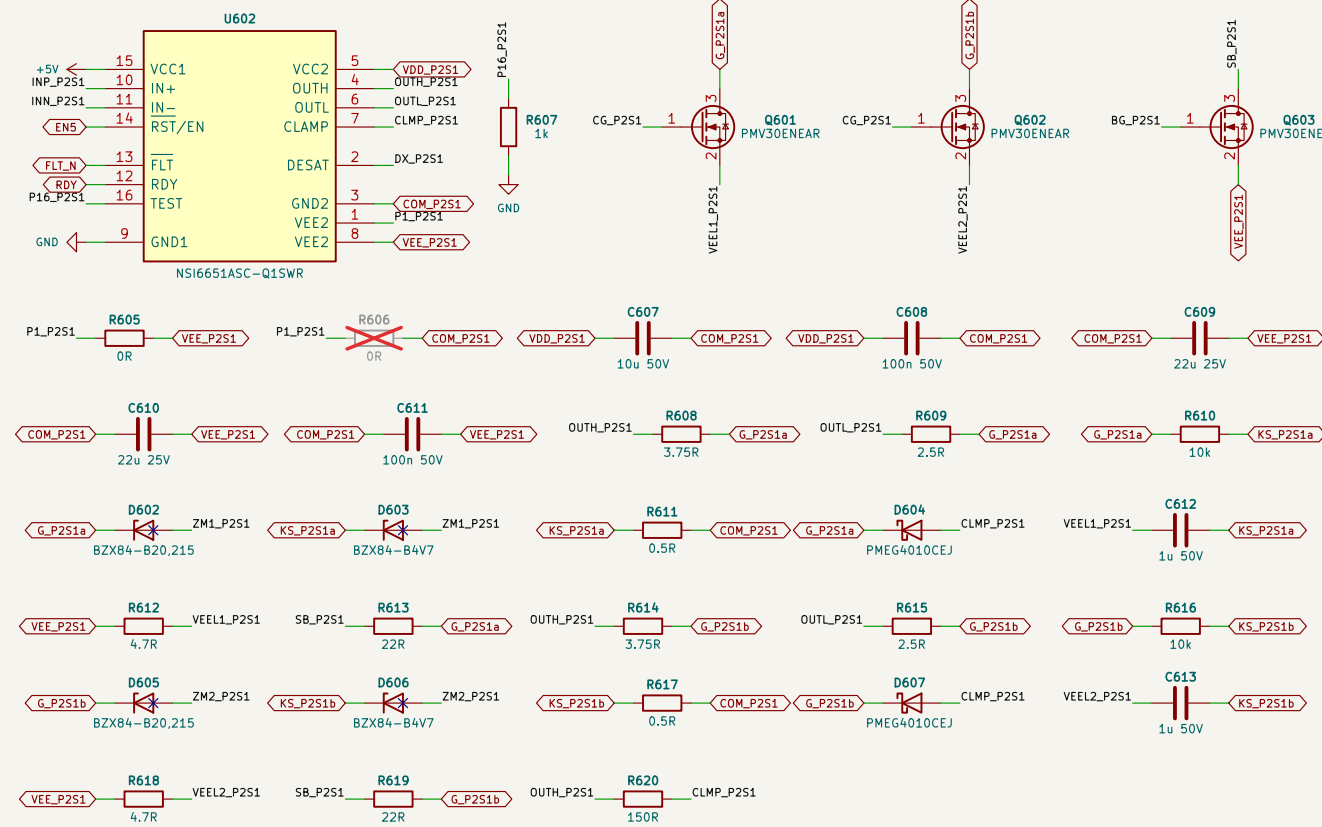


P2S1: LOGIC SIDE



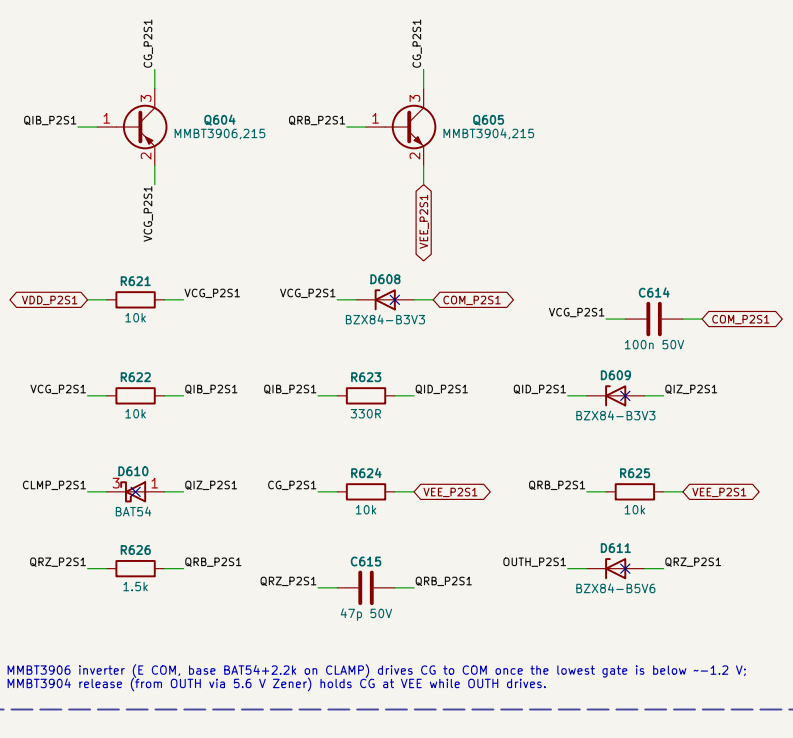
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S1: DRIVER AND 2 GATES



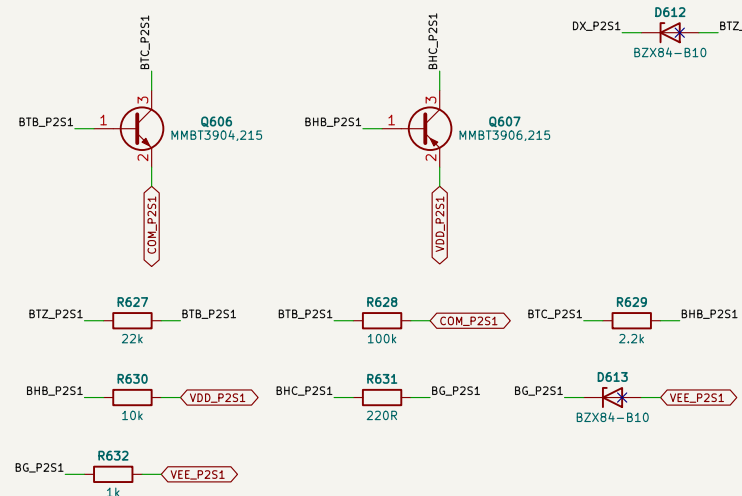
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Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S1: MILLER CLAMP CONTROL



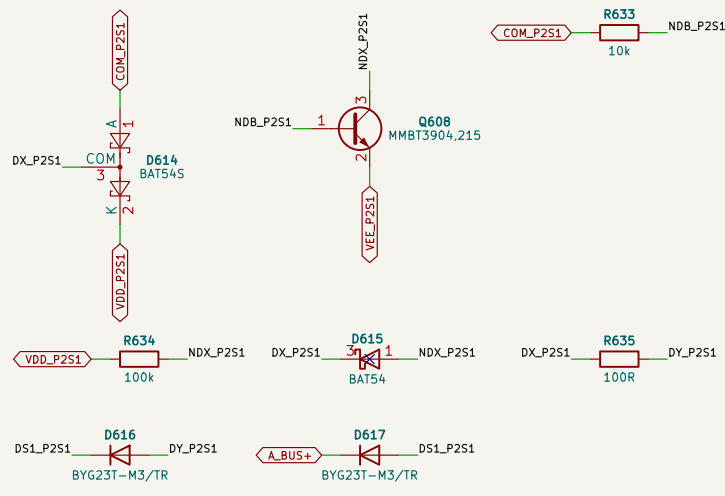
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V:
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S1: SHORT-CIRCUIT BOOSTER



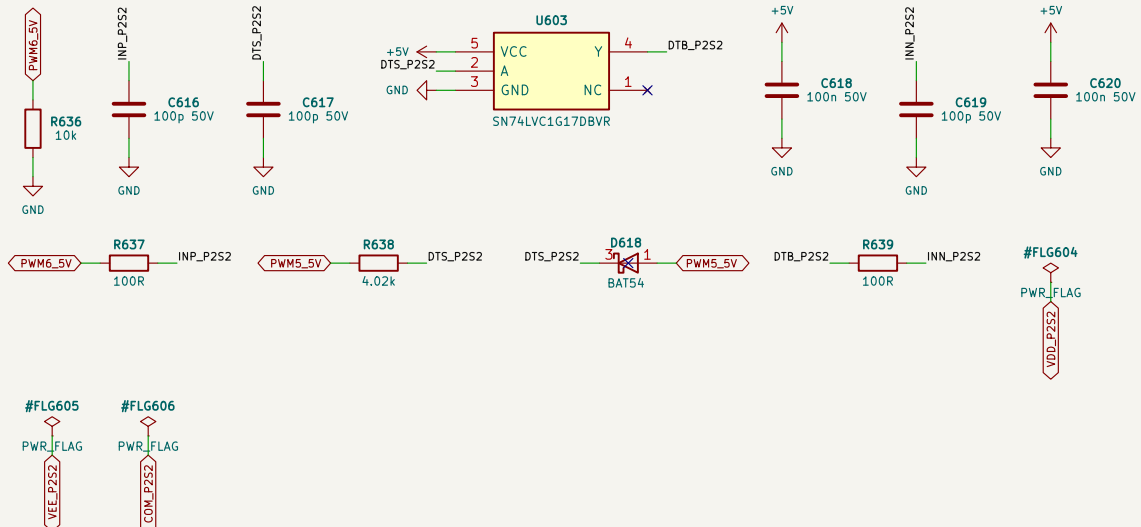
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V:
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P2S1: DESAT



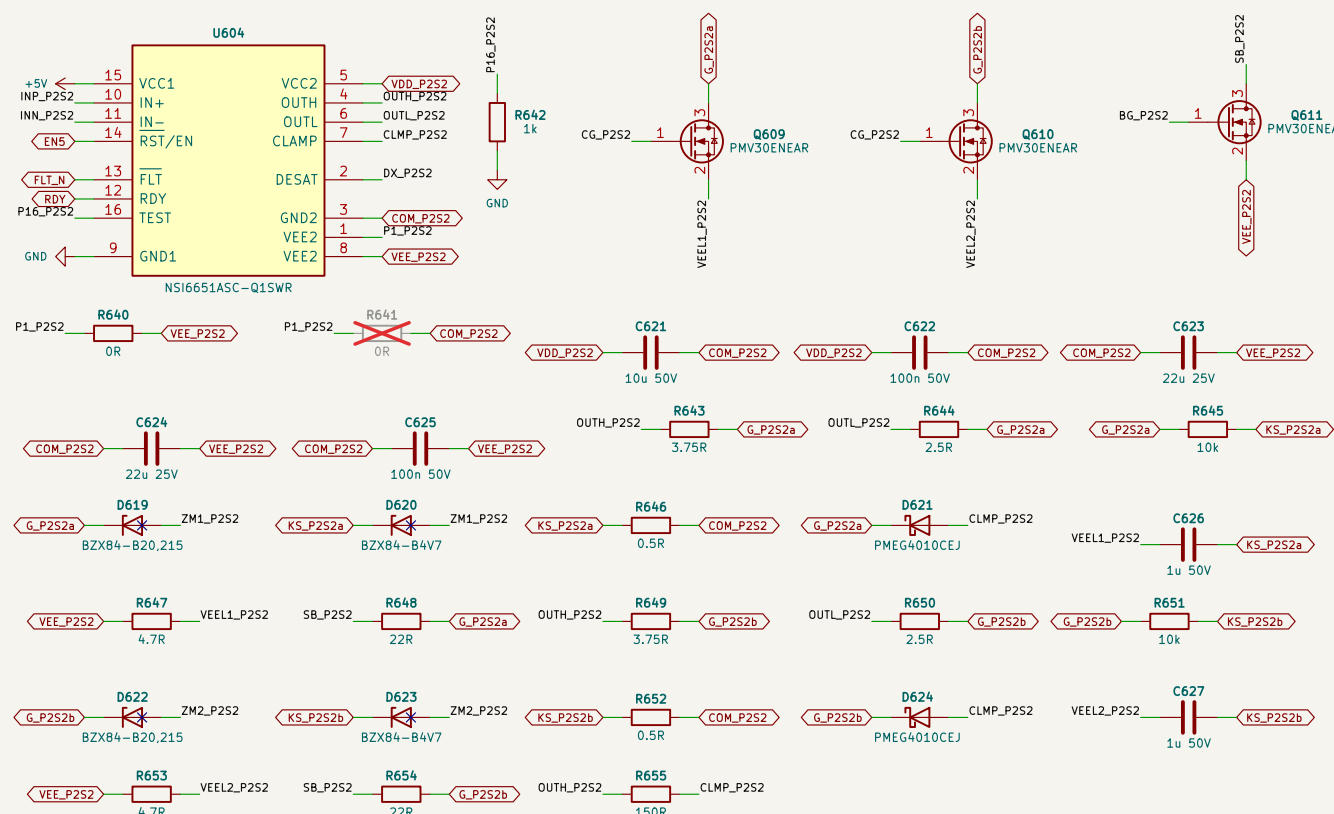
DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Calk footprint DMP.

P2S2: LOGIC SIDE



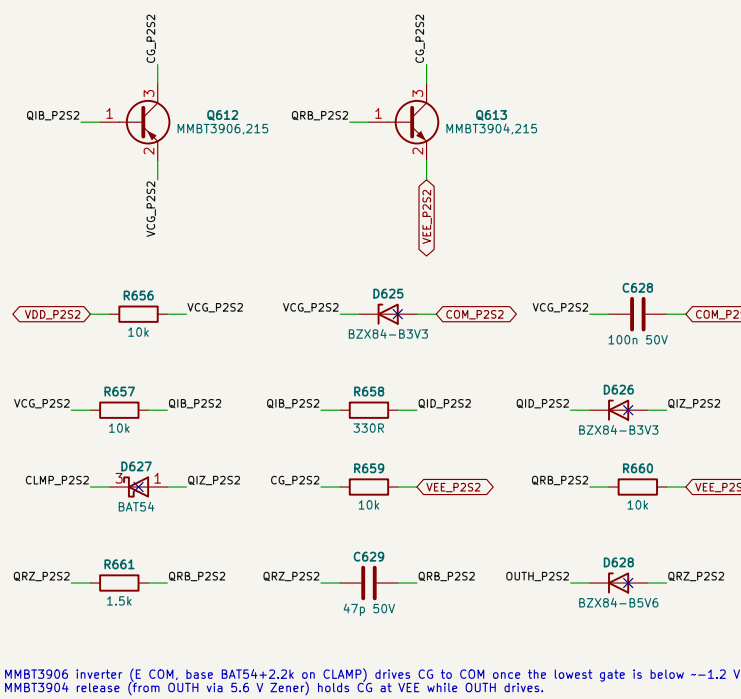
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S2: DRIVER AND 2 GATES



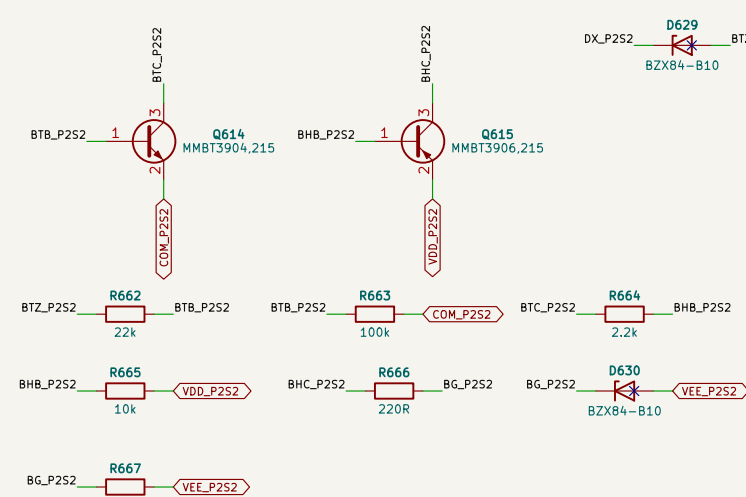
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S2: MILLER CLAMP CONTROL



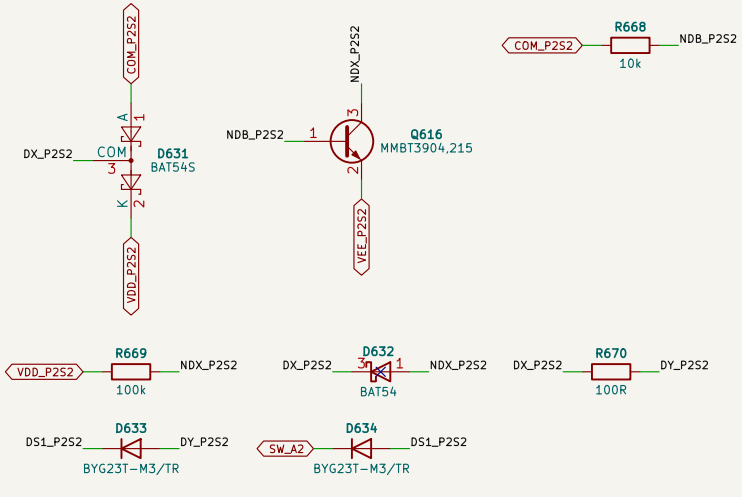
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V:
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V:
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (4k + FET gates).

P2S2: DESAT

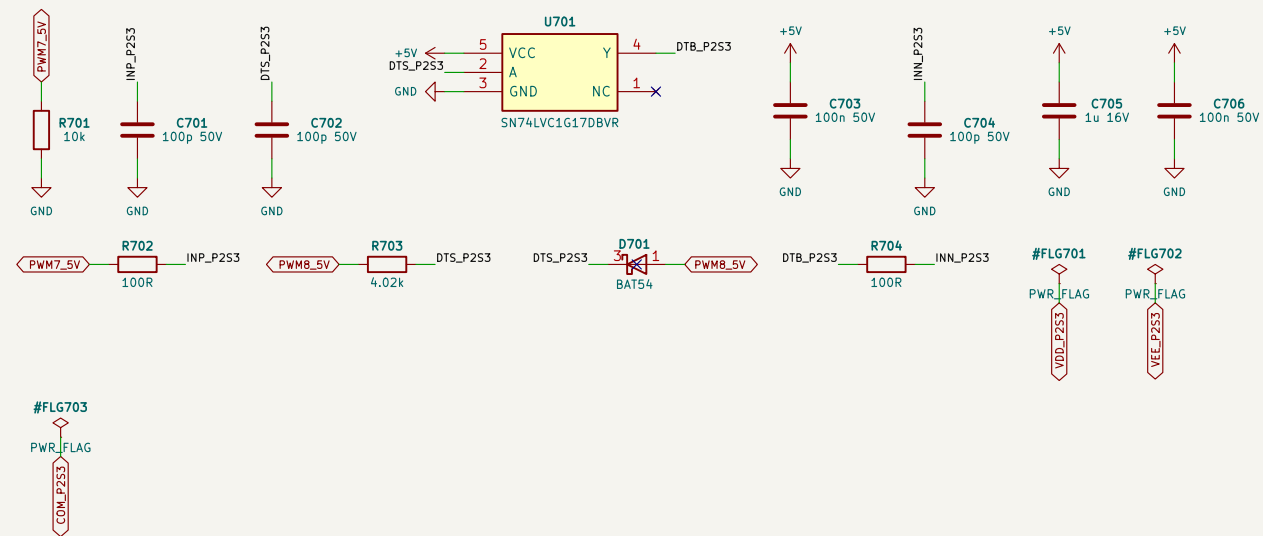


DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Calk footprint DMP.

LEG A: LAYOUT RULES

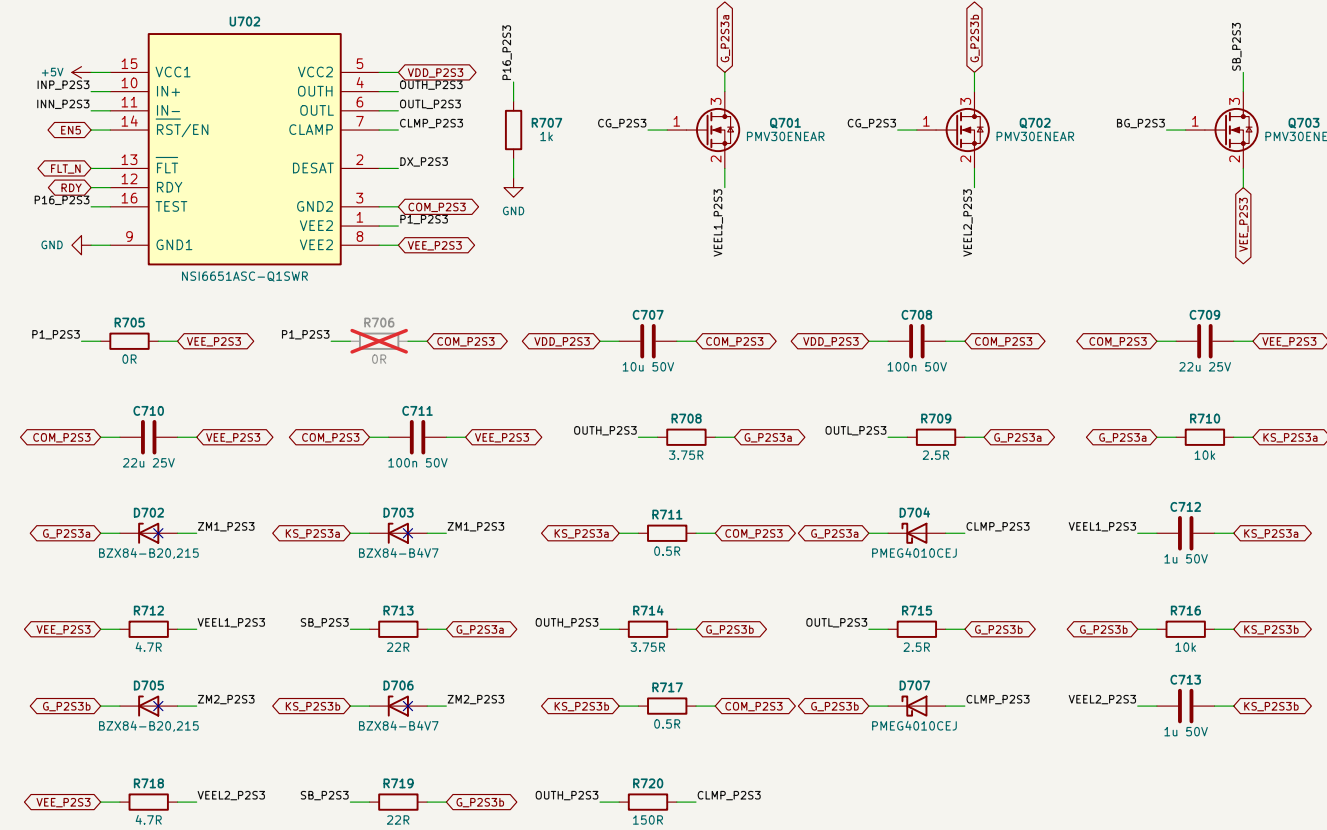
Gate and Kelvin of each device as coupled pairs of equal length (+/-10%) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to P01 (IC-22). Miller hold (nagpipe, groy): die +1.4 V vs VGS(n) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

P2S3: LOGIC SIDE



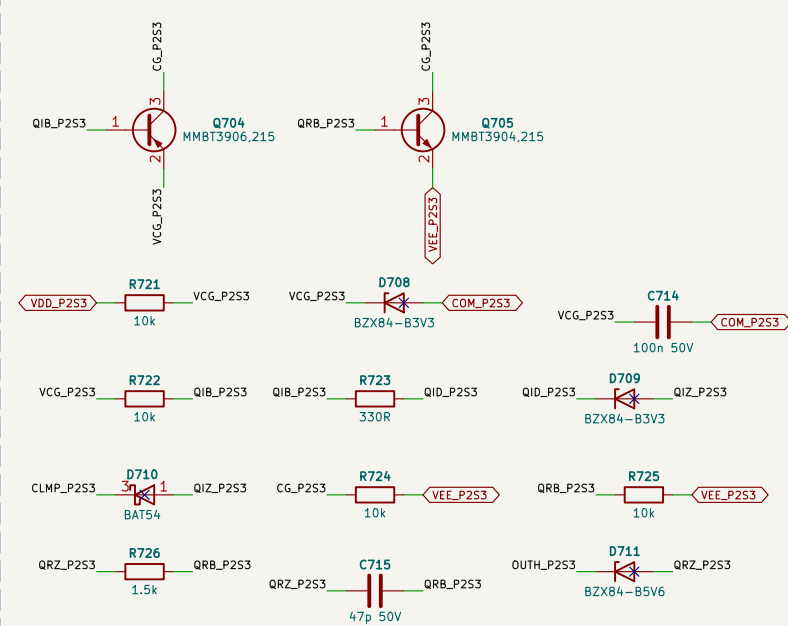
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S3: DRIVER AND 2 GATES



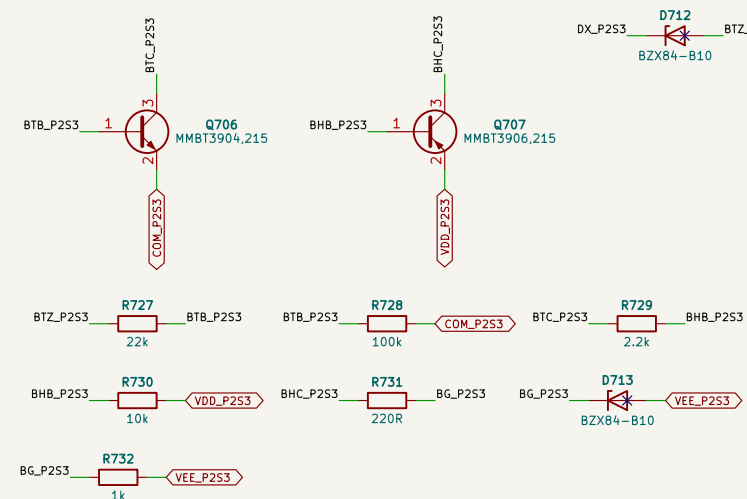
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S3: MILLER CLAMP CONTROL



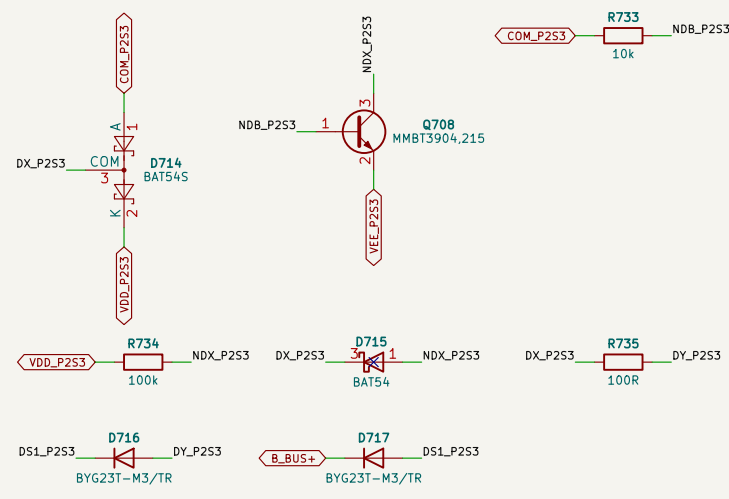
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S3: SHORT-CIRCUIT BOOSTER



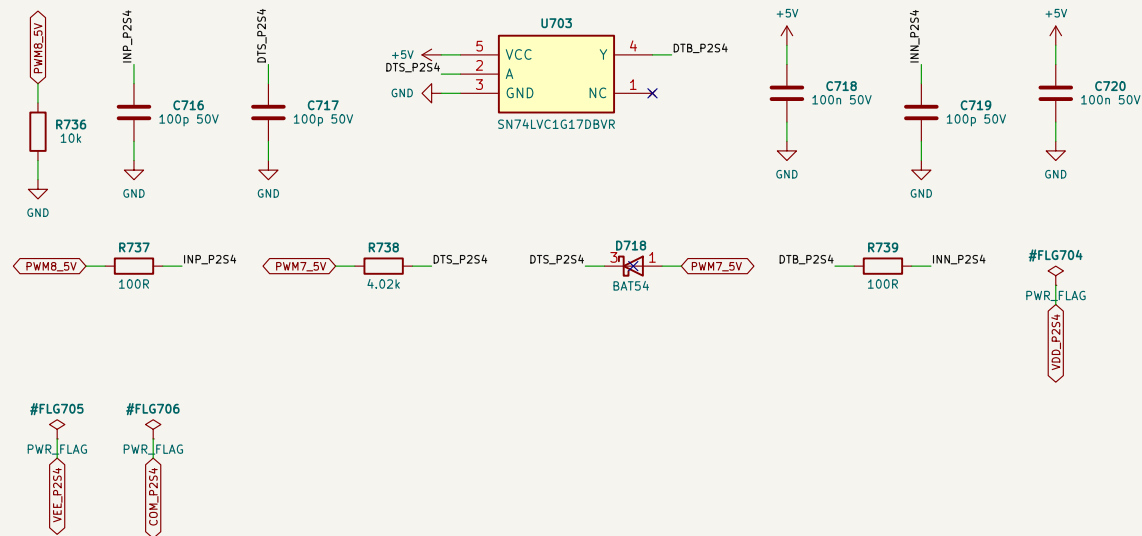
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P2S3: DESAT



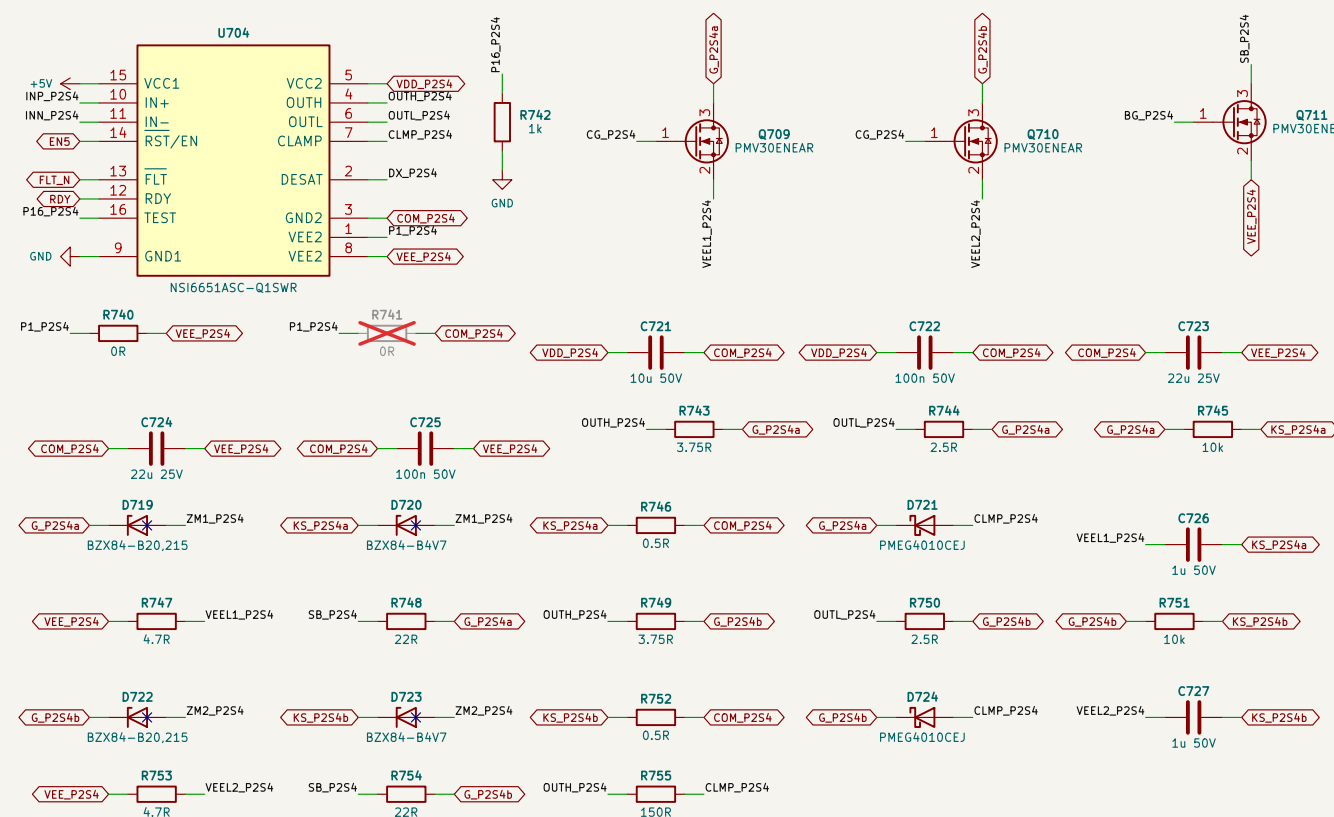
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P2S4: LOGIC SIDE



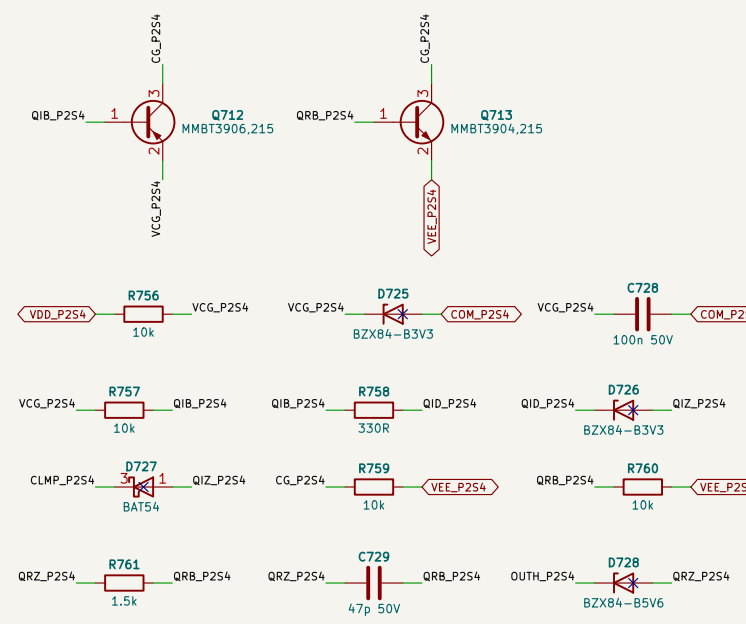
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P2S4: DRIVER AND 2 GATES



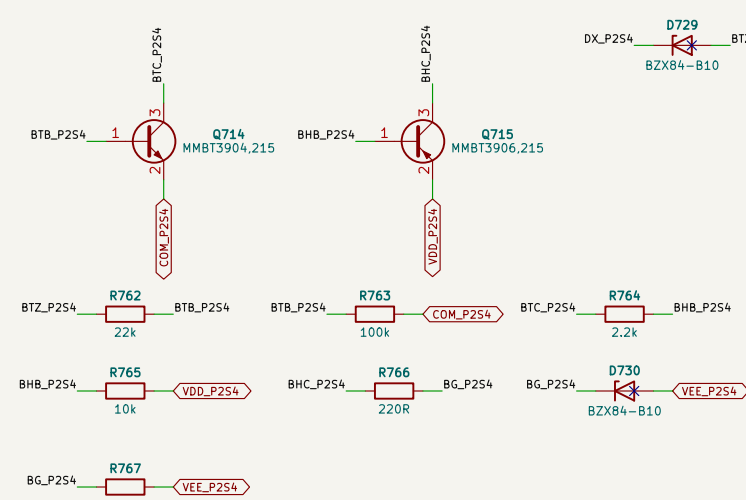
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P2S4: MILLER CLAMP CONTROL



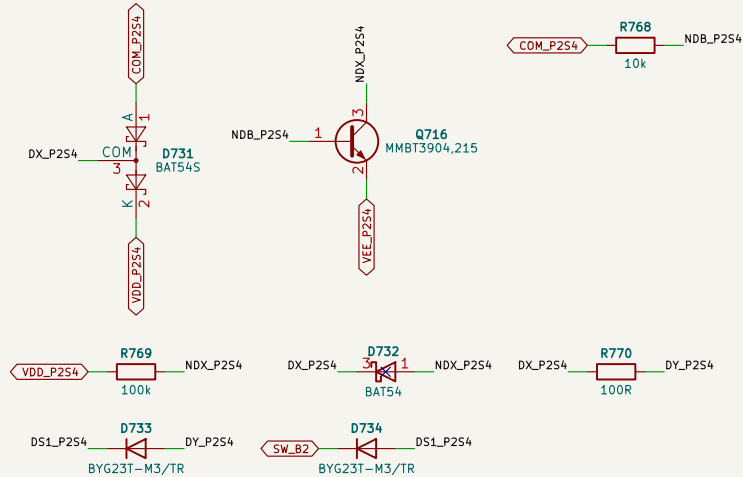
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P2S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P2S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length ($\pm 10\%$) from the COM star; Ron, Roff, Kelvin R, 10k. Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop ≤ 1 nH. DESAT string ≥ 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die ± 1.41 V vs VGS(th) min 1.94 V at 175 C – BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 8
Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basis

Sheet: /07 Phase 2 leg B gate drive (S3, S4)/

File: 07_ph2_gdb.kicad_sch
Title: DV, DV/D, 4-layer board - Phase 2 las B auto drive (S3, S4)

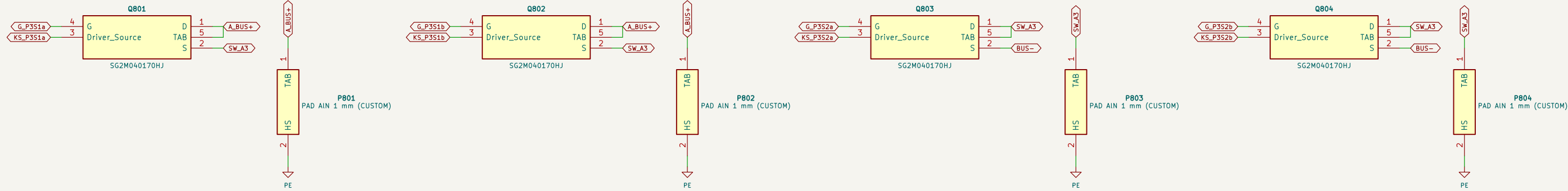
Title: PV-PWR-4 power board	
Size: A2	Date: 2026-10-05

KiCad E.D.A. 10.0.6

Rev: A2

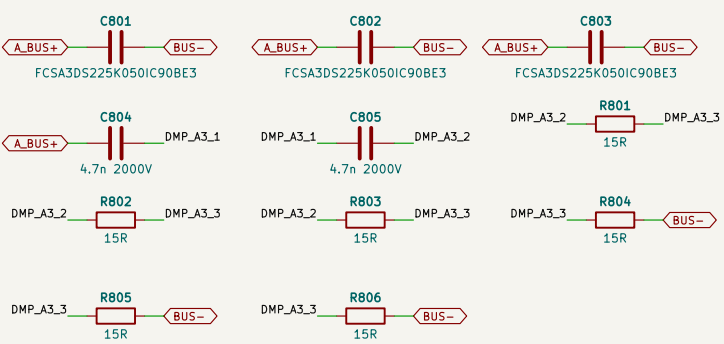
Id: 8/26

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



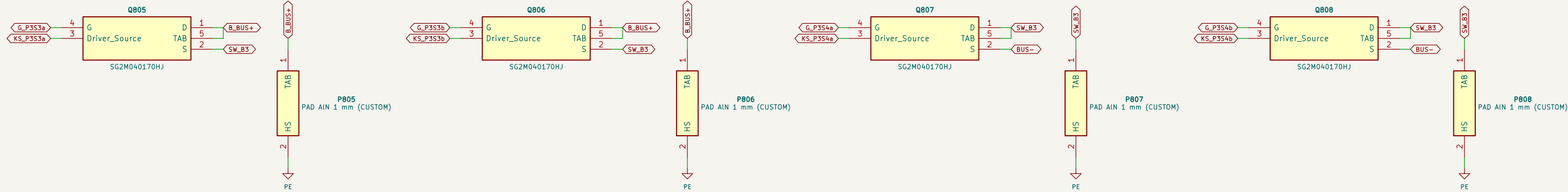
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



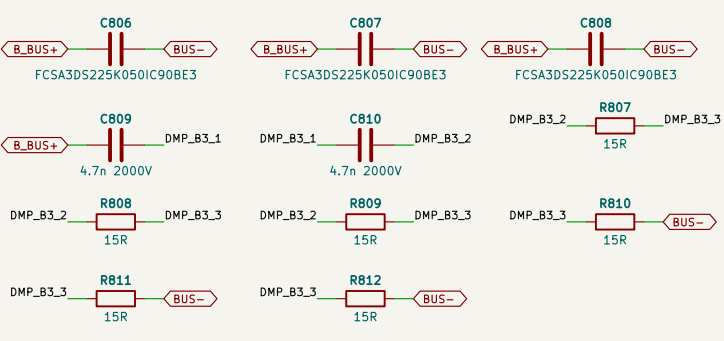
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



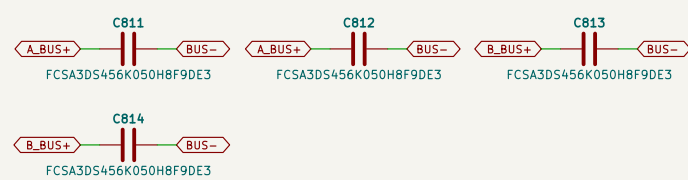
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



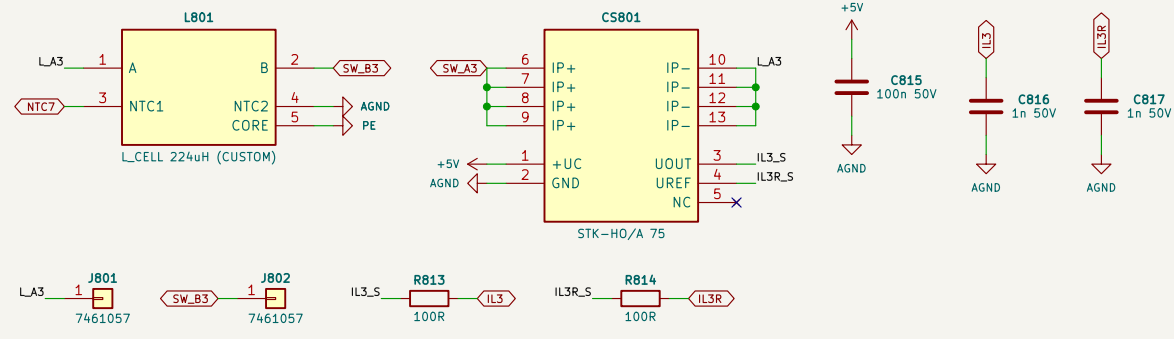
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



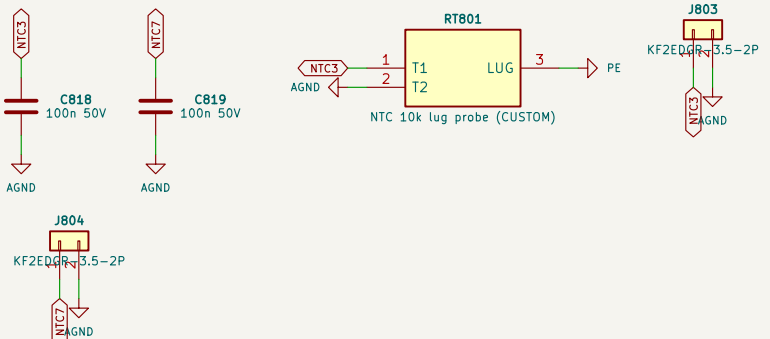
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



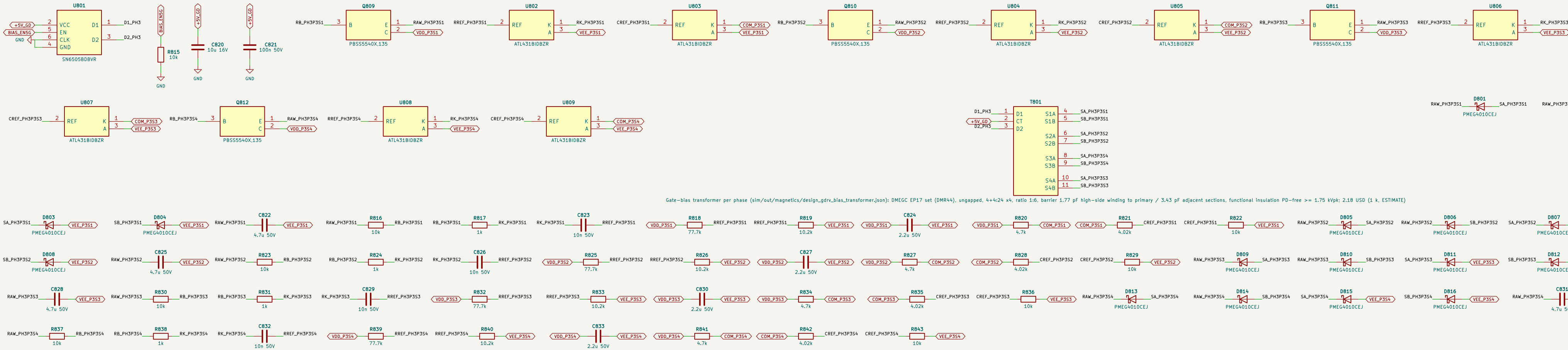
Lead A from SW_A3 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out; + = SW_A -> SW_B) to stud
LA3; lead B on stud SW_B3, IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mF at the source, IL3R = the sensor's own Uref (pin 4), 100R +/- 1 mF; PV-CTL references its trip
window to it. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



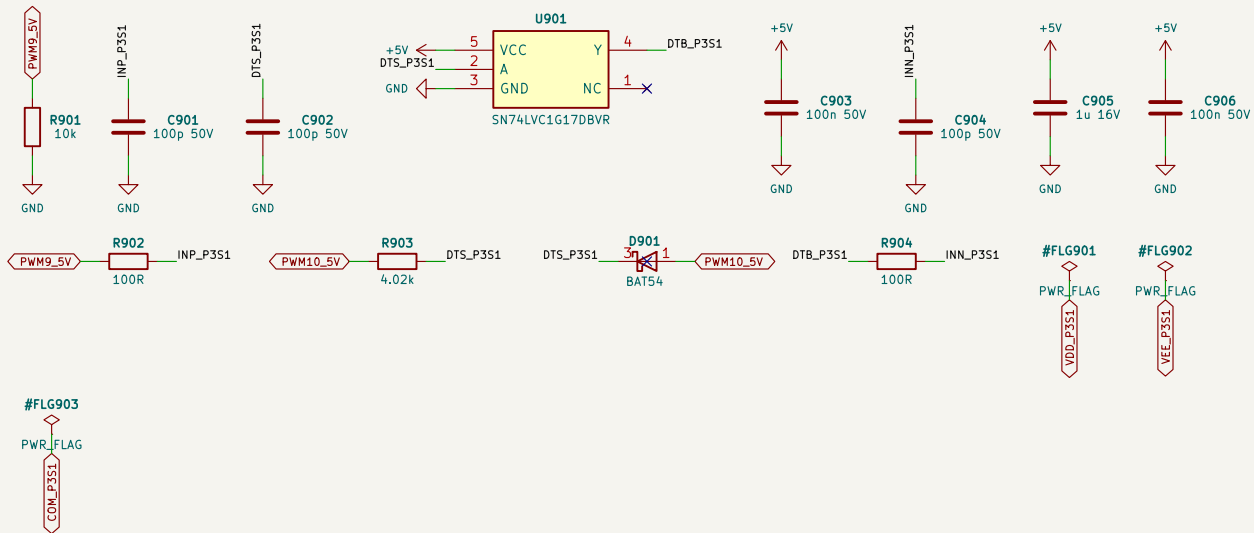
NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC7 = inductor winding NTC (functional to the winding), 100 mF at the header against switch-node noise.

PH3: GATE BIAS (PER PHASE)



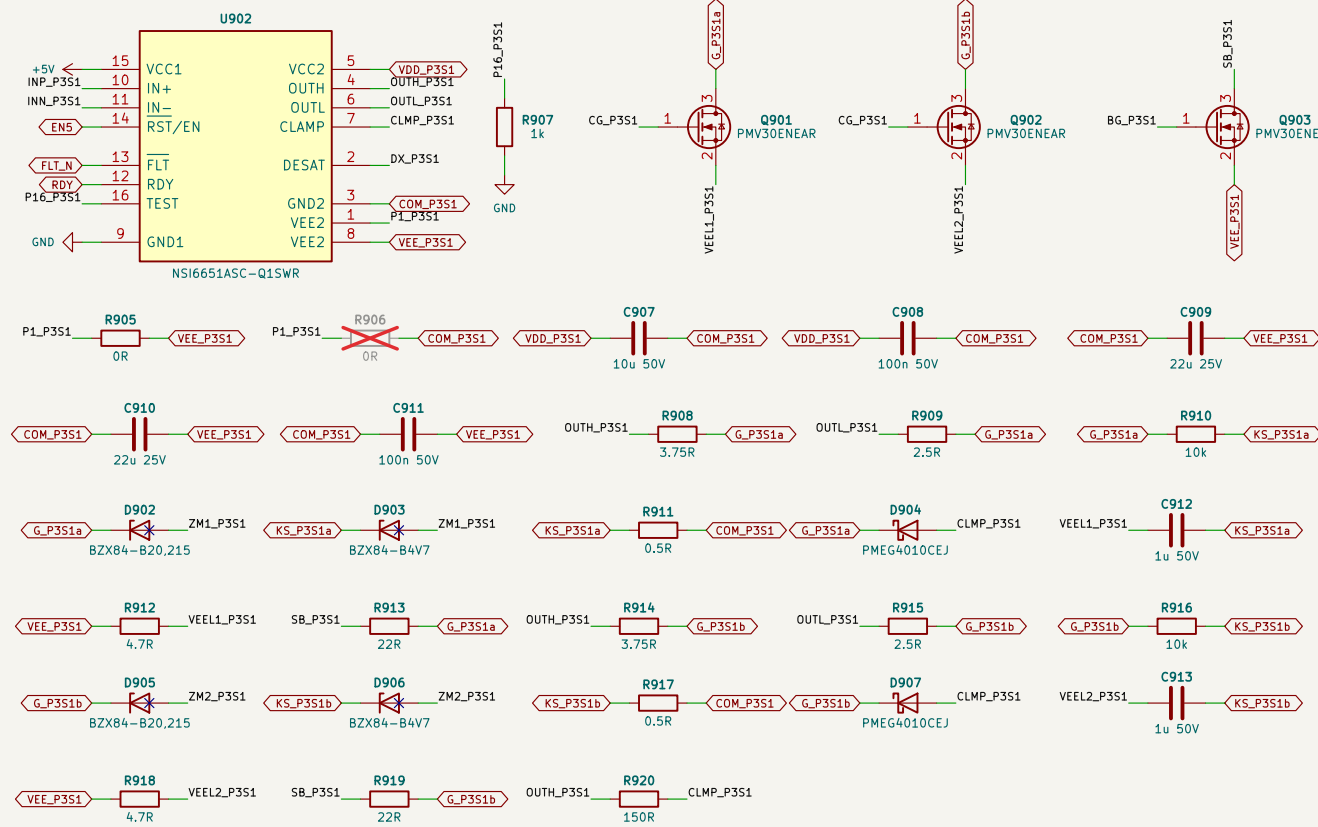
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T_BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



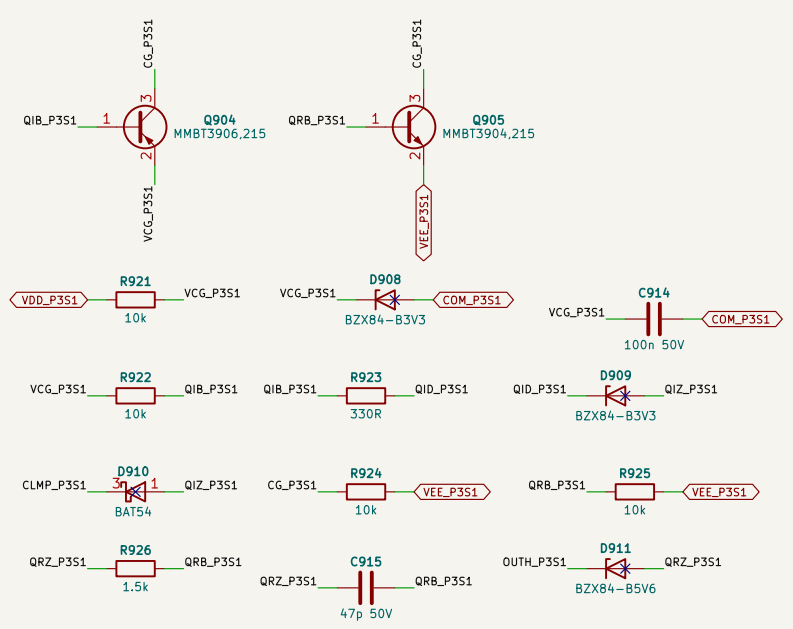
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



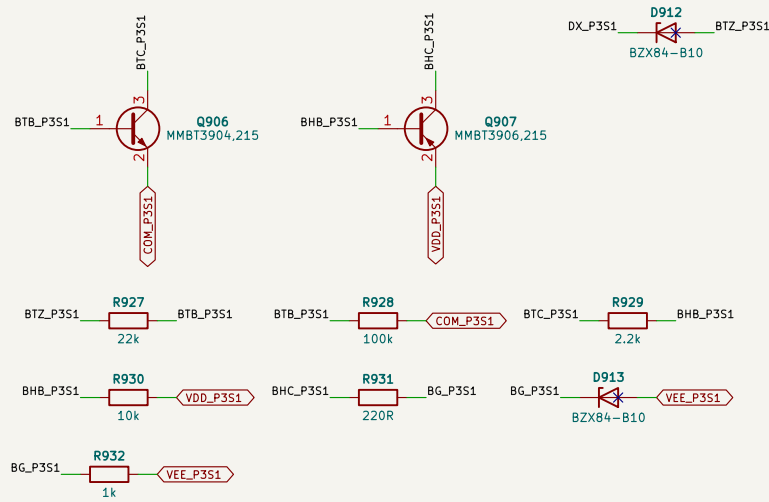
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



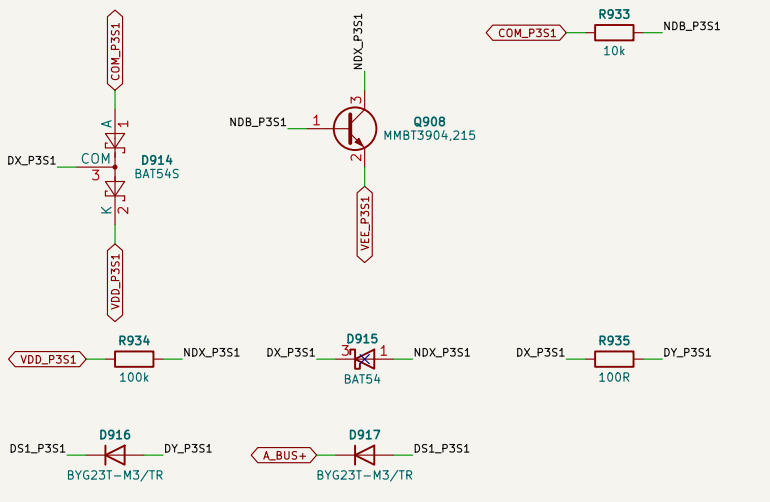
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MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



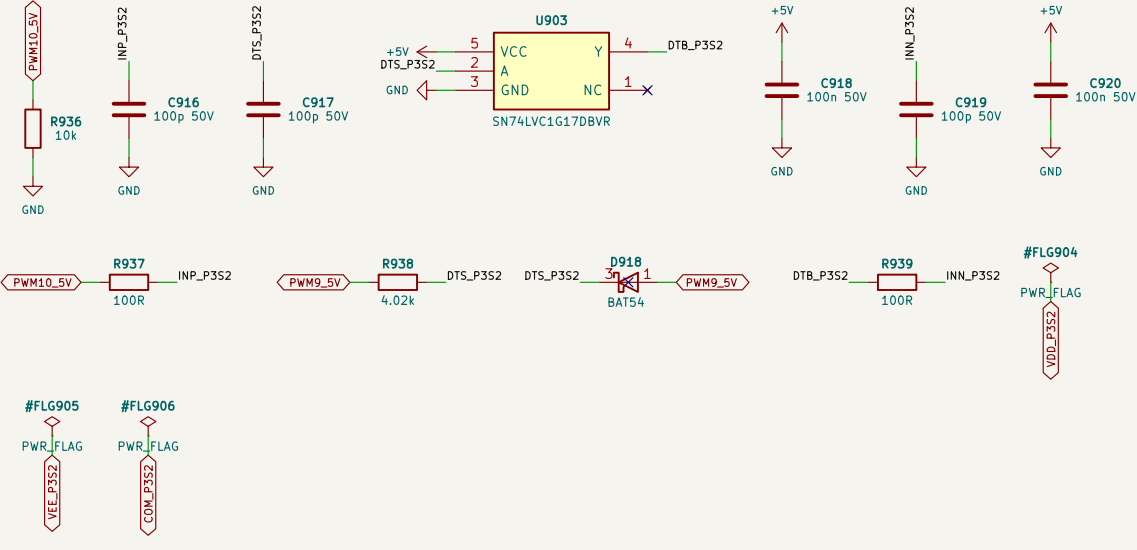
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per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (A + FET gates).

P3S1: DESAT



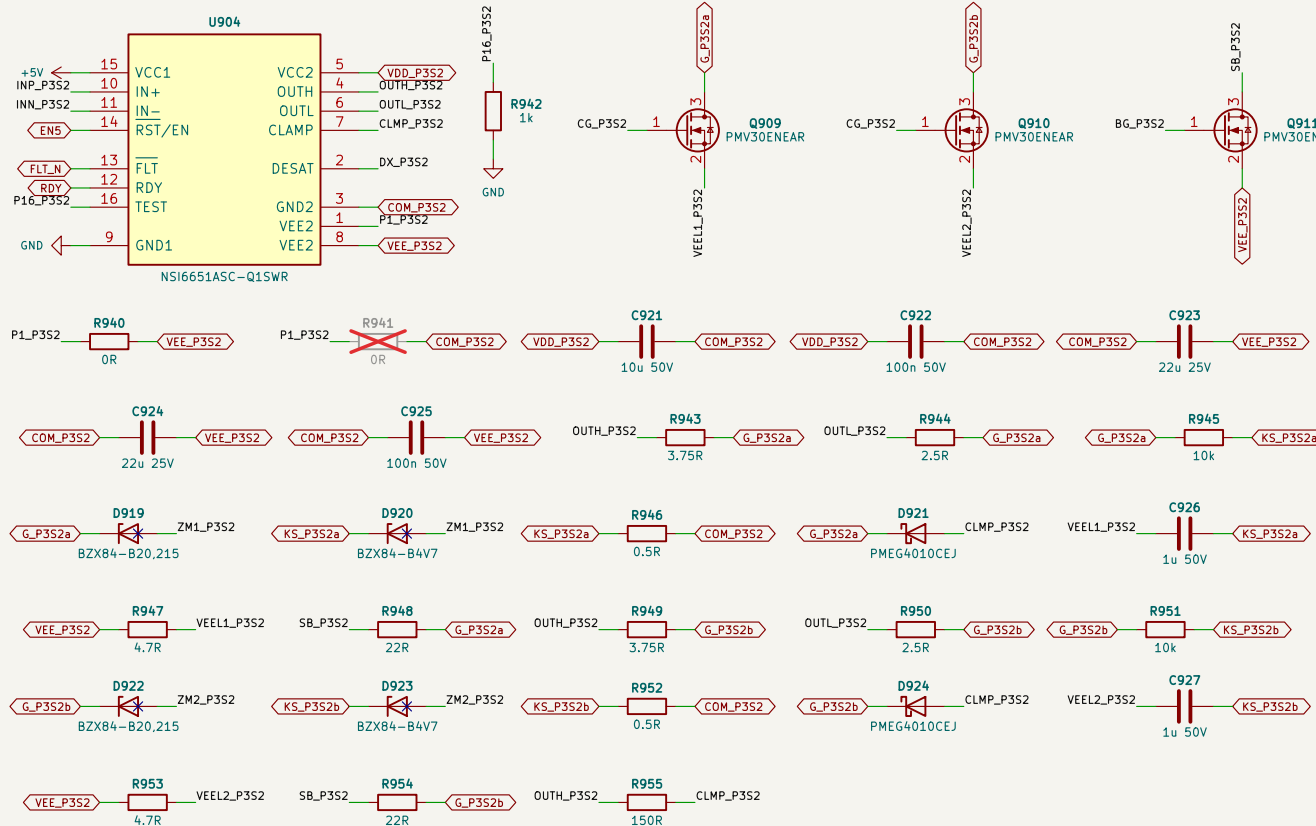
DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rs = 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

P3S2: LOGIC SIDE



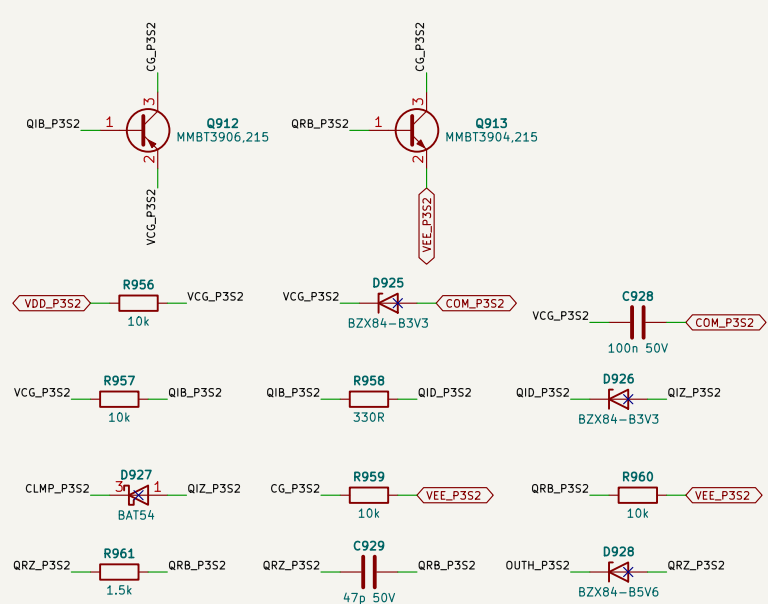
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



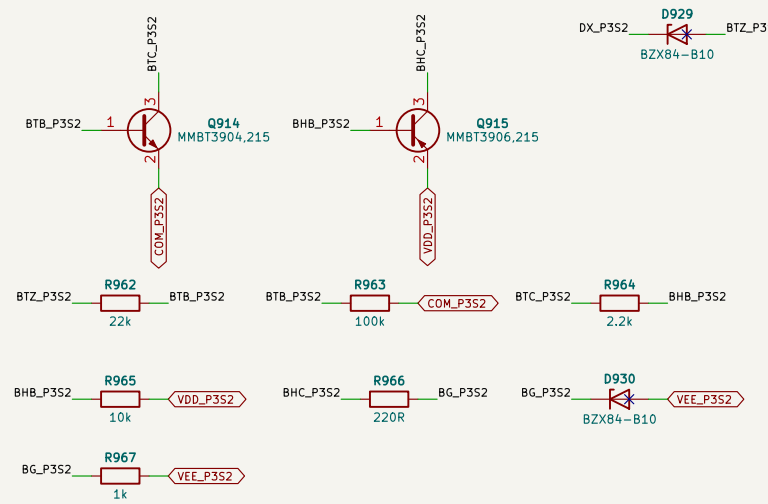
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30E9EA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



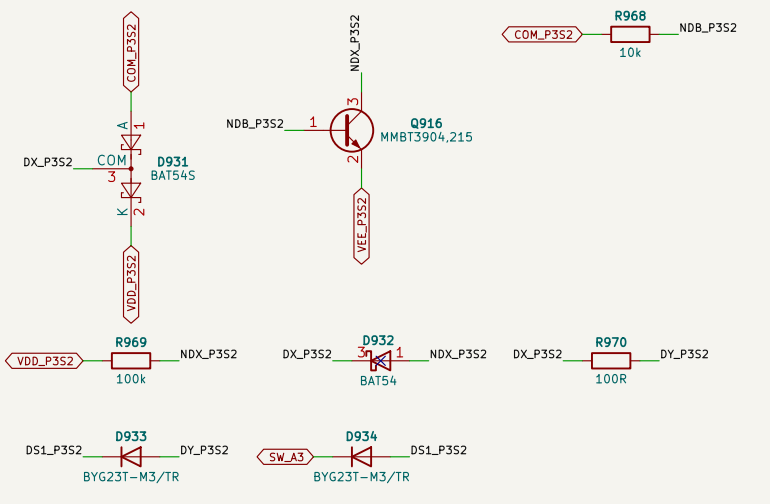
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V:
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P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V:
per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S2: DESAT

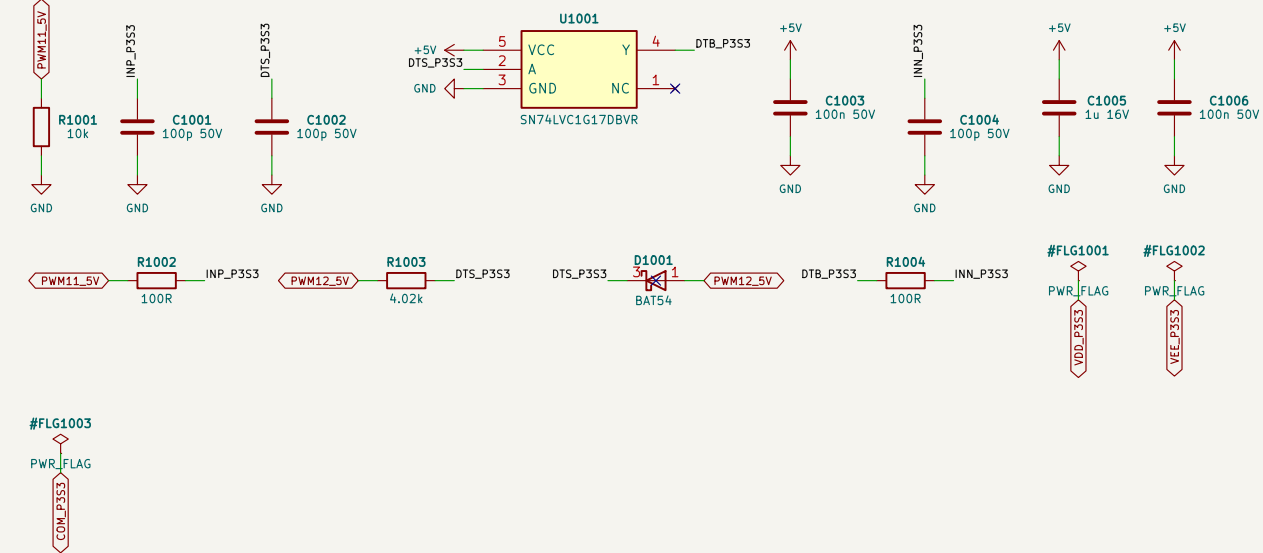


DESAT pin (Internal 500 uA blanking source, trip 8.5-10.0 V) -> Rs = 2 x BYG23T-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cbik footprint DNP.

LEG A: LAYOUT RULES

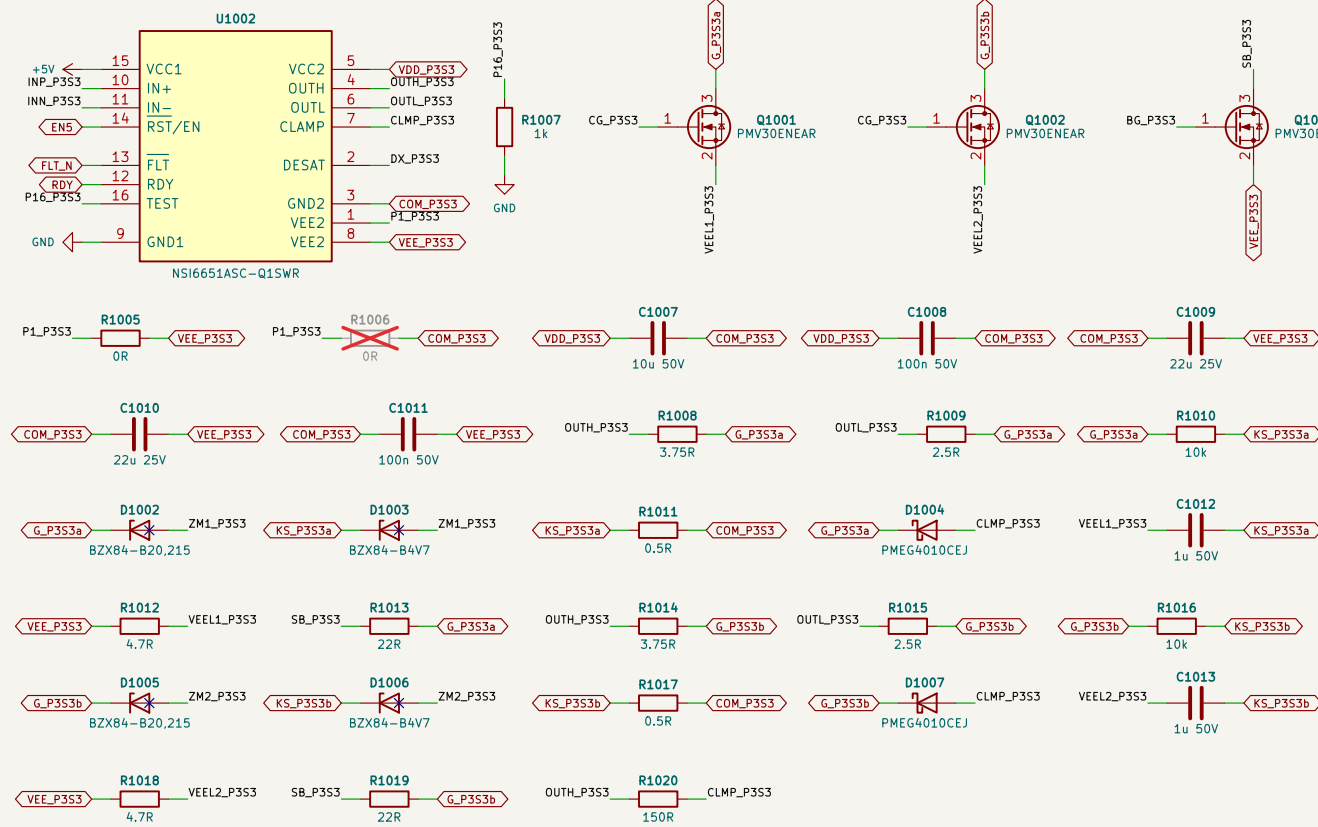
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P3S3: LOGIC SIDE



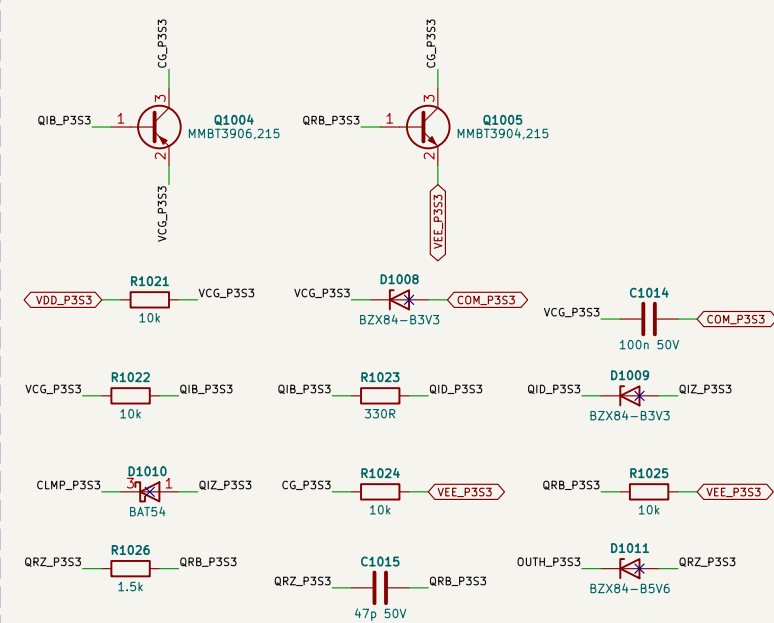
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



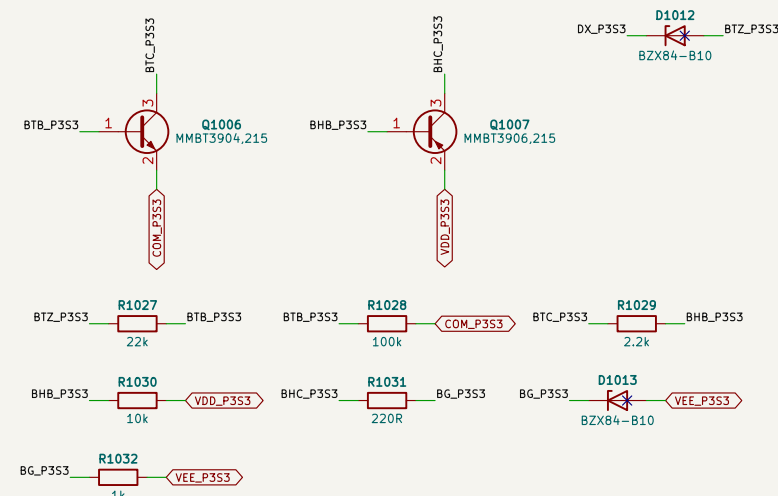
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENE4) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



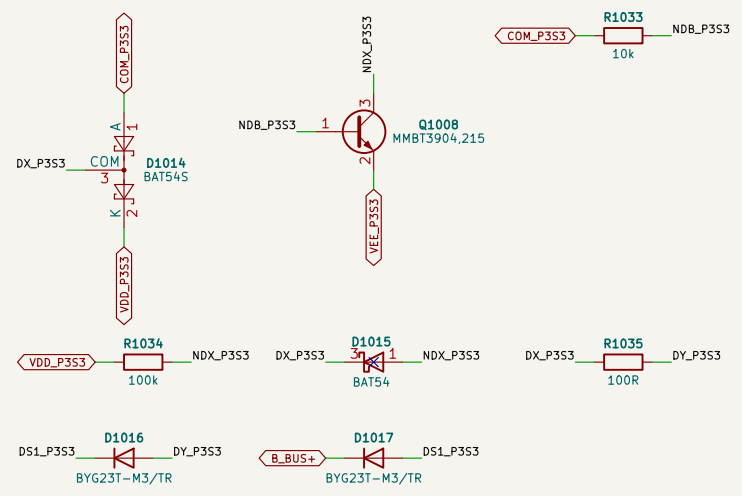
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below ~ -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



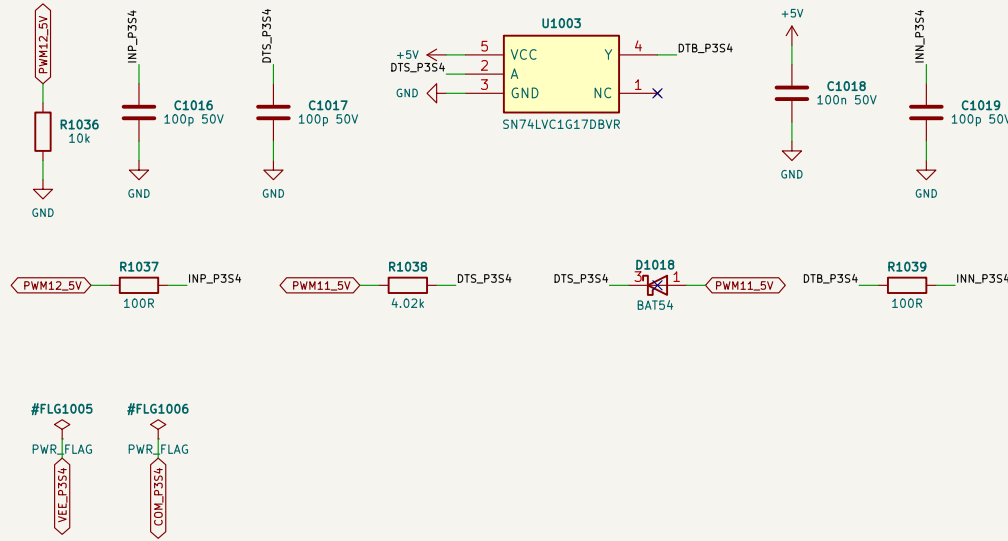
DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AO3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S3: DESAT



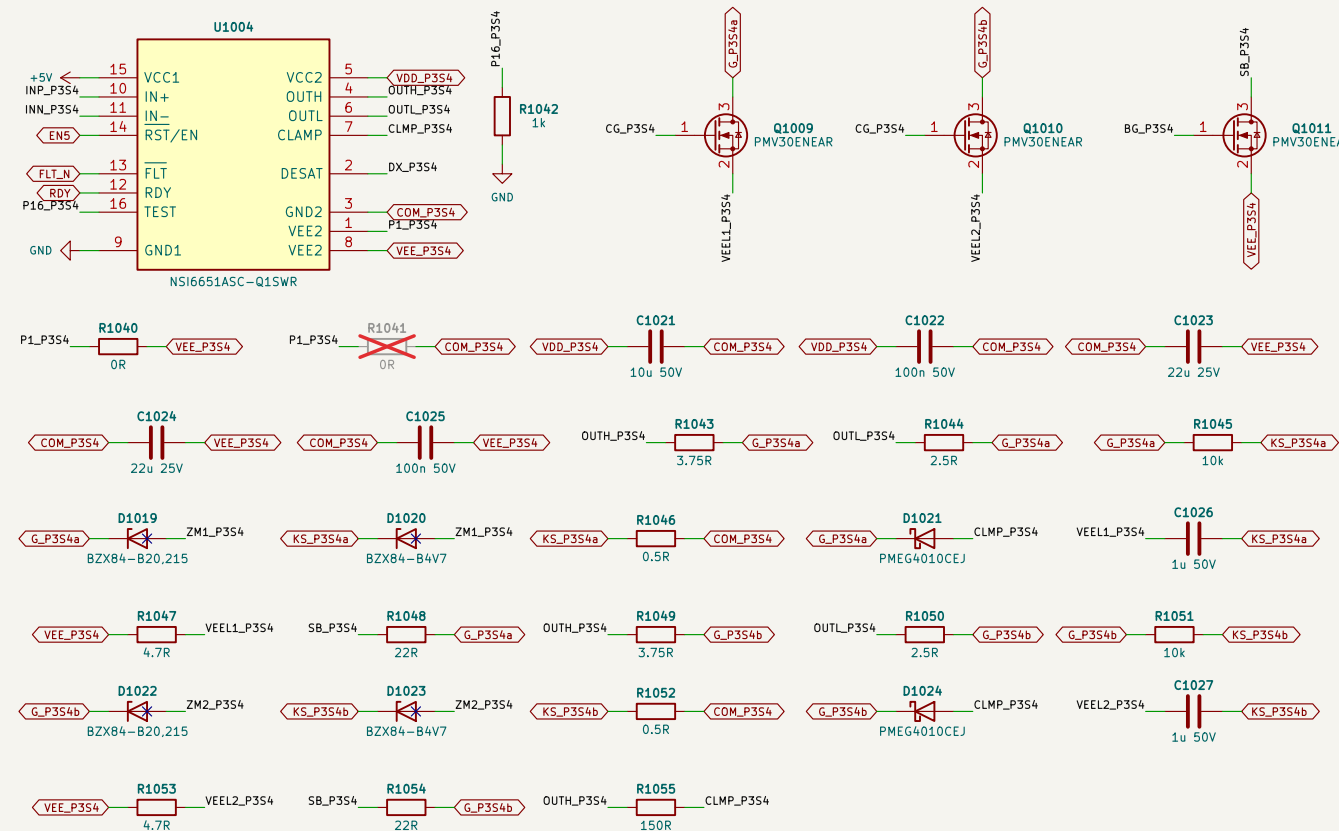
DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs = 2 x BYG23T–M3/TR (1300 V each) – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

P3S4: LOGIC SIDE



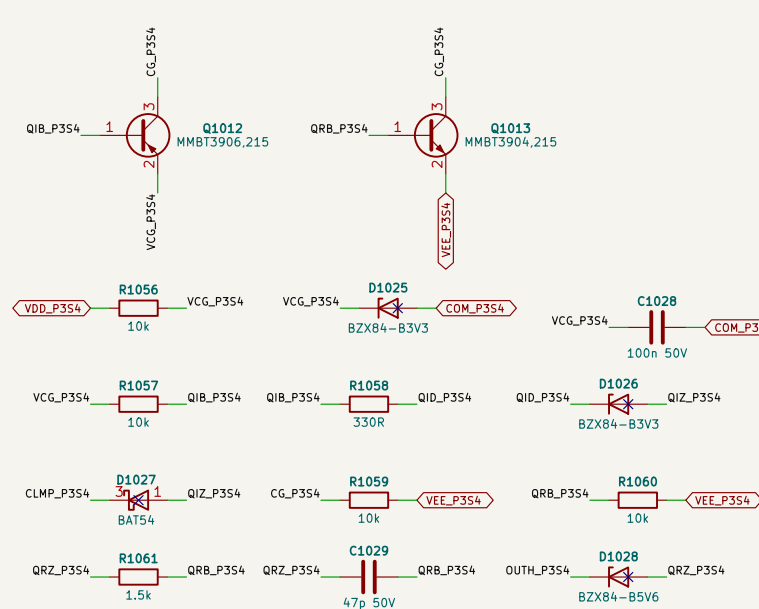
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



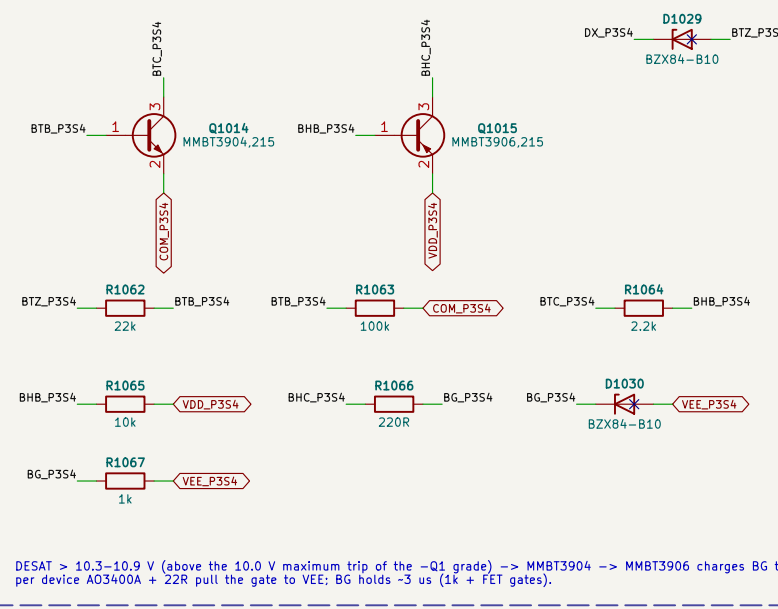
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



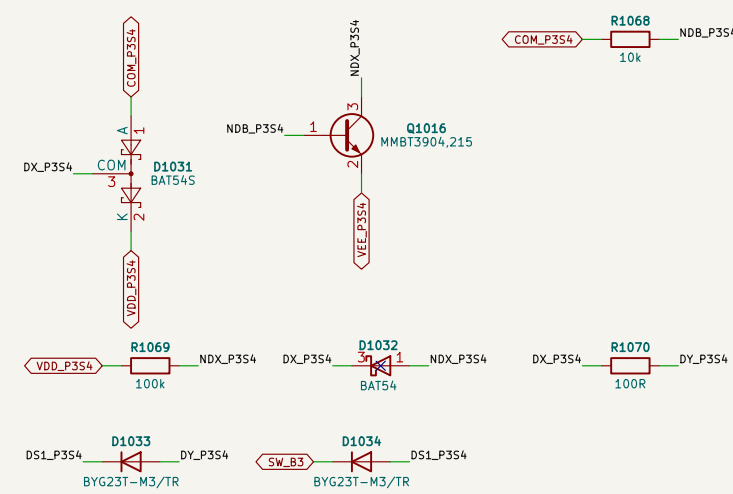
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device AO3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs = 2 x BYG23T–M3/TR (1300 V each) – drain.
BATS4S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 1.0k, Zeners at each device. Clamp FET + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (ngspice, gdrv): dte +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 14

Sheet 11
Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE base

Sheet: /10 Phase 3 leg B gate drive (S3, S4)/

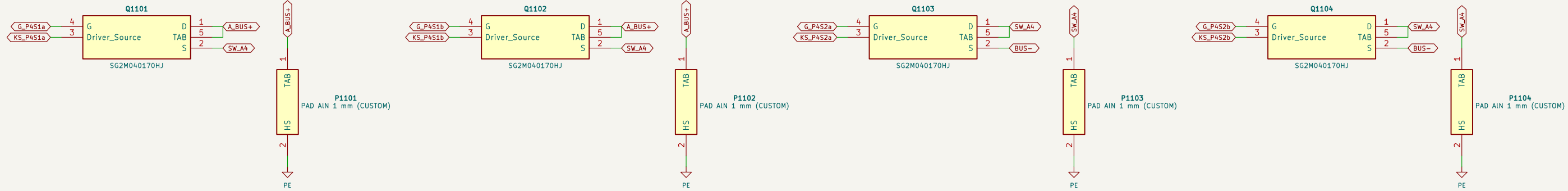
Title: PV-PWR-4 power board – Phase 3 leg B gate drive (S3, S4)

Size: A2	Date: 2026-10-05
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KiCad E.D.A. 10.0.6	Id: 11/26
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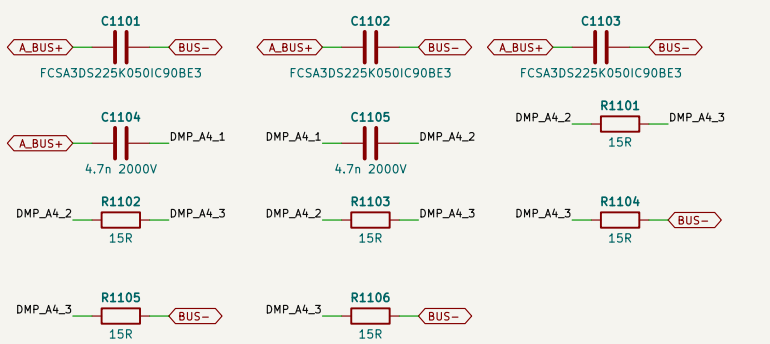
10

PHASE 4 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



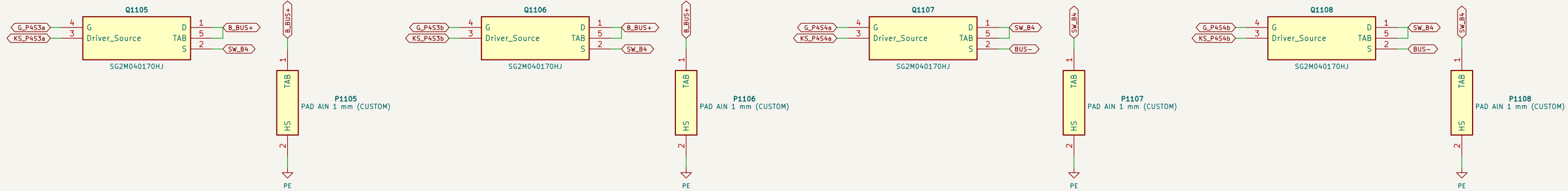
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG A DECOUPLING + RC DAMPER



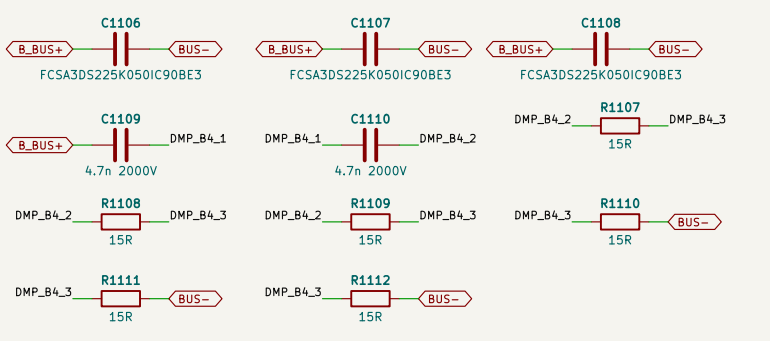
3 x FCSA3D5225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



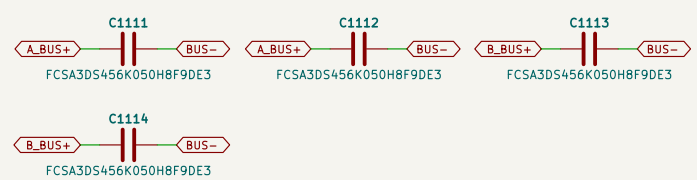
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AIN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4 equal drain / source busbar paths for the two devices of a switch.

PHASE 4 LEG B DECOUPLING + RC DAMPER



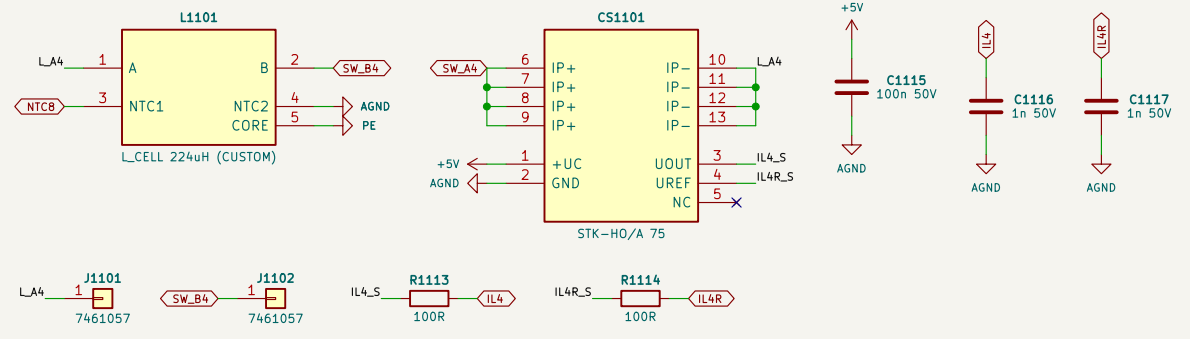
3 x FCSA3D5225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV COG in series, loop at the pins <= 3 nH.

PHASE 4 SHARE OF THE PORT BANKS



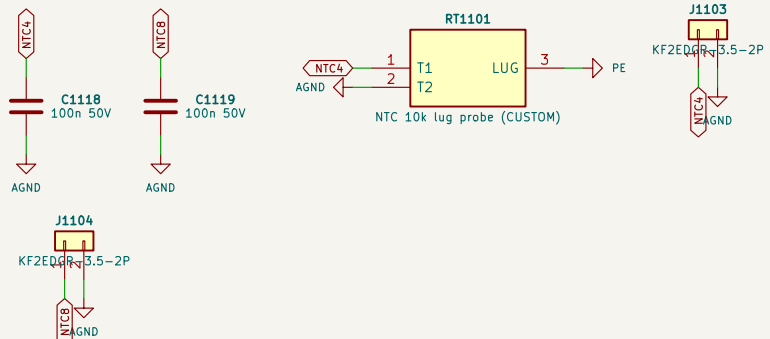
2 x FCSA3D5456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs; the banks of all phases form the
port A / port B DC links (8 / 8 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 4 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



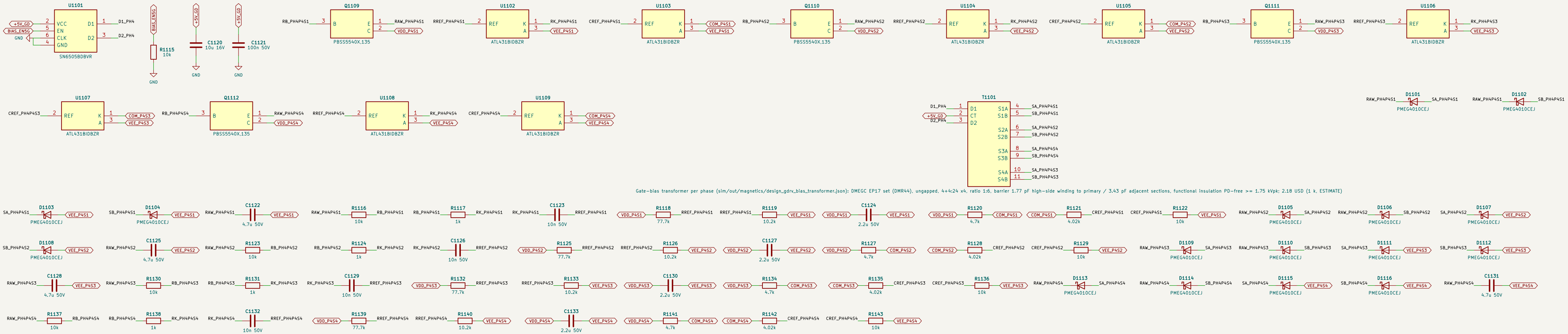
Lead A from SW_A4 passes the STK-HQ/A 75 U-primary (pins 6-9 in, 10-13 out) + SW_A -> SW_B to stud
L_A4; lead B on stud SW_B4; IL4 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V). 0.5-4.5 V over +/-187.5 A;
100R +/- 1 mV at the source; IL4R = the sensor's own Uref (pin 4). 100R +/- 1 mV; PV-CTL references its trip
window to it. Primary at the switch node; functional barrier (4 kV rms, 8 kV impulse).

PHASE 4 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC4 = heatsink section of phase 4; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (8-10).
NTC5 = inductor winding NTC (functional to the winding). 100 mV at the header against switch-node noise.

PH4: GATE BIAS (PER PHASE)



PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_{JN}- / A_{JN}-, bank A_{BUS}-. No fuse, no precharge; array lsc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec_lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side A_{IN}+ / A_{IN}-, bank A_{BUS}+. No fuse, no precharge; array I_{sc} ≤ 1.25 x rating.
Contactor opens ≤ 0.97 kA, held above (port_spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

TYT25751 x 3 (615 VDC, thermal link + monitor lead): MOV1 +/M, MOV2 -/M, MOV3 M/PE.T. Up. eff 3793 V pole-P1 and pole-pole at in 3 kA (calc., <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 3A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line): RB1 = 1 at rest above 200 V = varistor network open -> CTRL inhibits/stops.

Sheet: /14 Port A power path (pv)/		
File: 14_leanA_power.kicad_sch		
Title: PV-PWR-4 power board - Port A power path (pv)		
Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 15/26

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 16 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Port A sensing and interlocks/ File: 15_leanA_ctrl.kicad_sch		
Title: PV-PWR-4 power board – Port A sensing and interlocks		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 16/26	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 17
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Port A coil drives/
File: 16_leanA_drive.kicad_sch

Title: **PV-PWR-4 power board – Port A coil drives**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

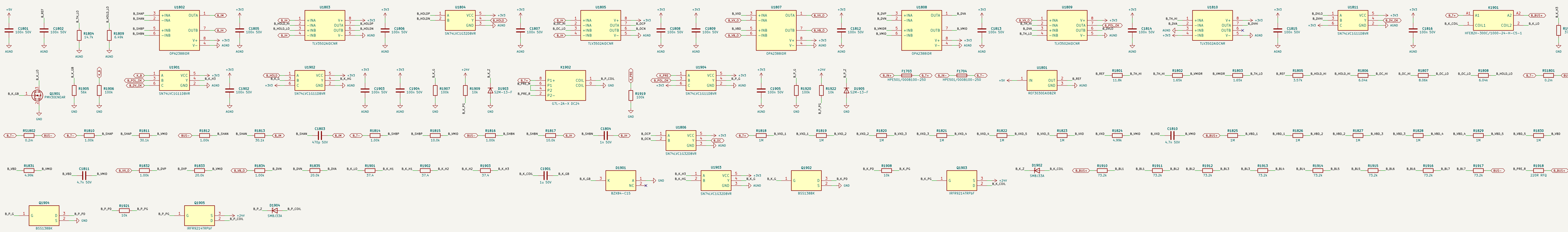
Id: 17/26

Terminal side B_{LN+} / B_{LN-}, bank B_{BUS+}. HPE501 250 A aR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

Terminal side B_{LN+} / B_{LN-}, bank B_{BUS+}. HPE501 250 A aR in both poles (breaks 1.25-50 kA only).
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

TVT25751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/M, MOV2 -/M, MOV3 M/PE_T, U_{g,eff} 3793 V pole-PE and pole-pole at in 3 kA (calc., $u = 4.0 \text{ kV} \Rightarrow$ reinforced 6 kV / basic 4 kV, D=032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three Units $\Rightarrow$ CNY658 $\Rightarrow$ RB1 (sum on the ID line); RB1 = 1 at rest above 200 V = varistor network open $\Rightarrow$ CTRL inhibits/stops.

C1801 C1802



HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

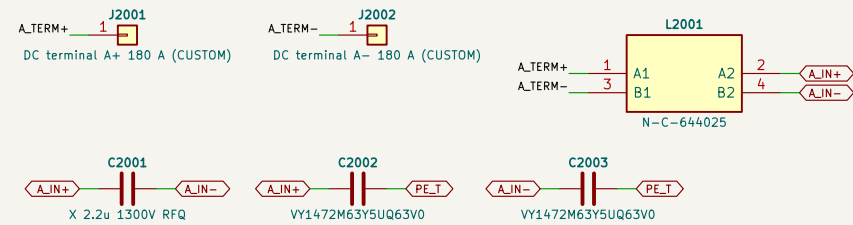
	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()		
Sheet 19		
Schematics authored by Chinmoy Bhuyan		
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /17 Port B sensing and interlocks/		
File: 17_leanB_ctrl.kicad_sch		
Title: PV-PWR-4 power board – Port B sensing and interlocks		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 19/26	

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

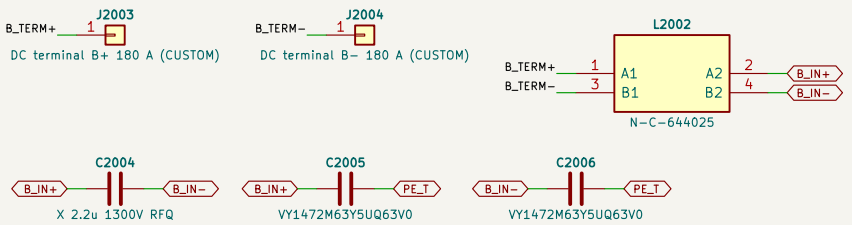
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 20 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /18 Port B coil drives/ File: 18_leanB_drive.kicad_sch		
Title: PV-PWR-4 power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 20/26	

PORT A TERMINALS AND EMI FILTER



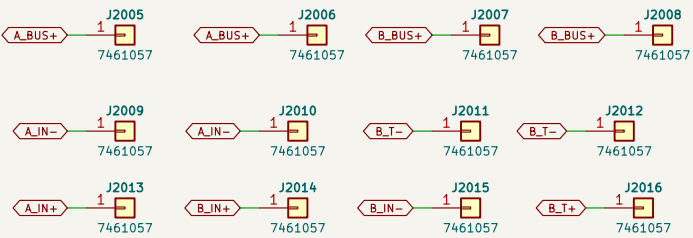
PV: chassis terminals A_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



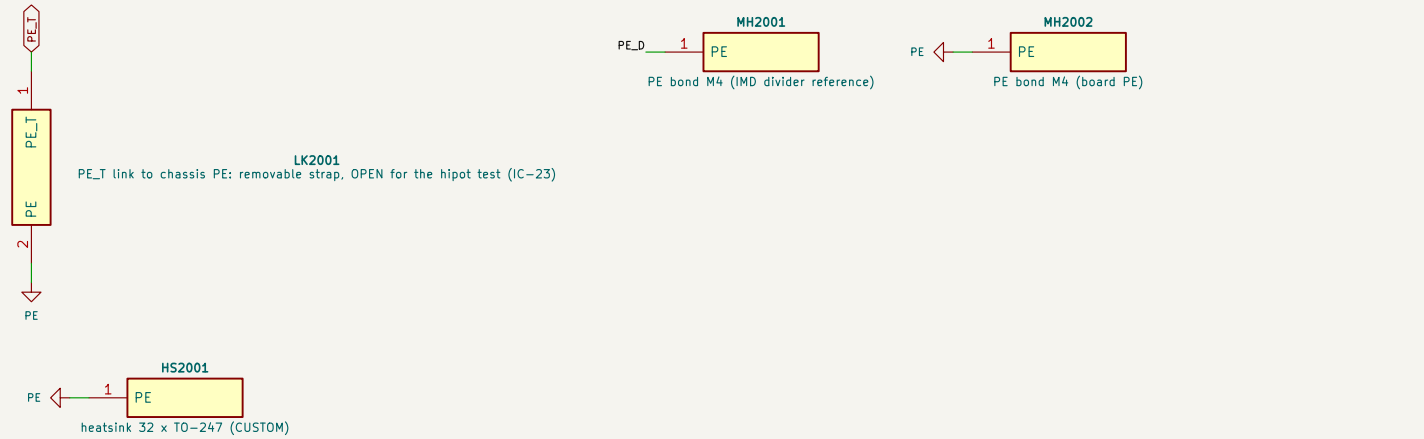
battery: chassis terminals B_TERM+/- (180 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1). X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



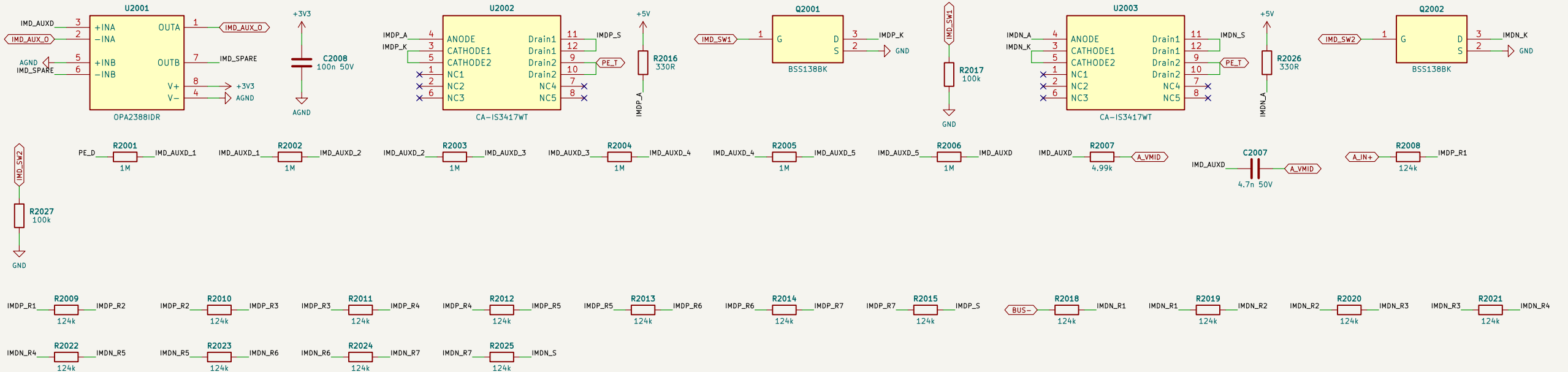
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



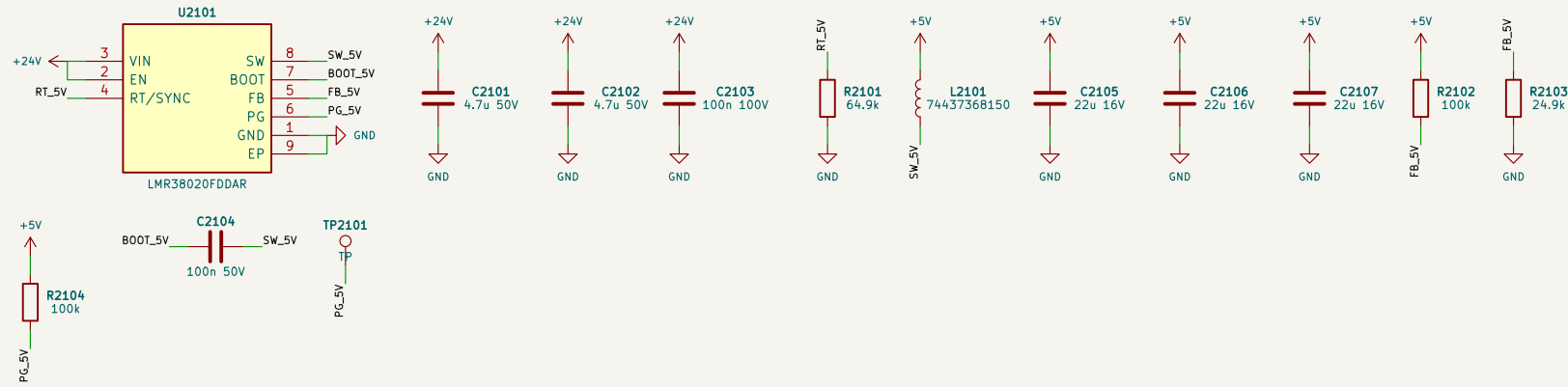
PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



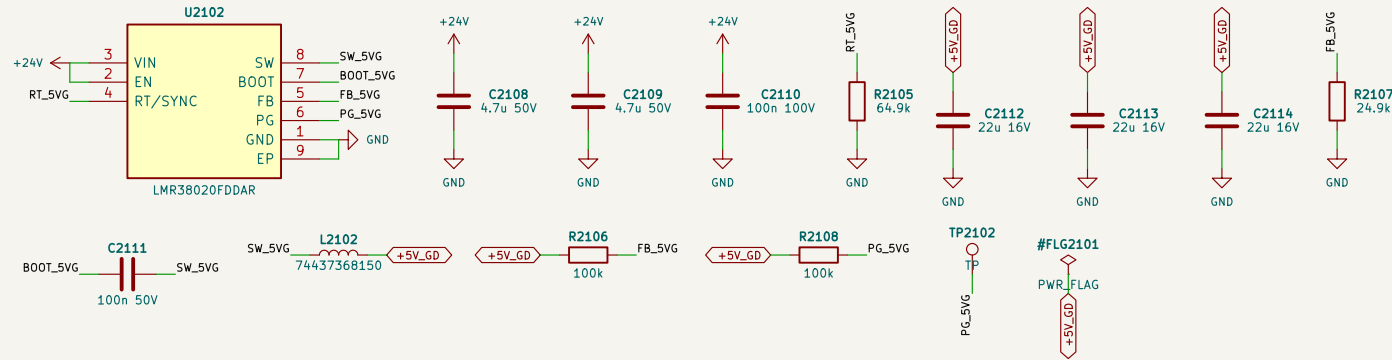
IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



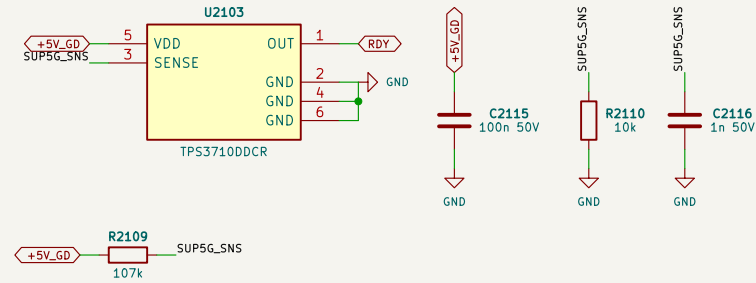
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1–2, ctrlC2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



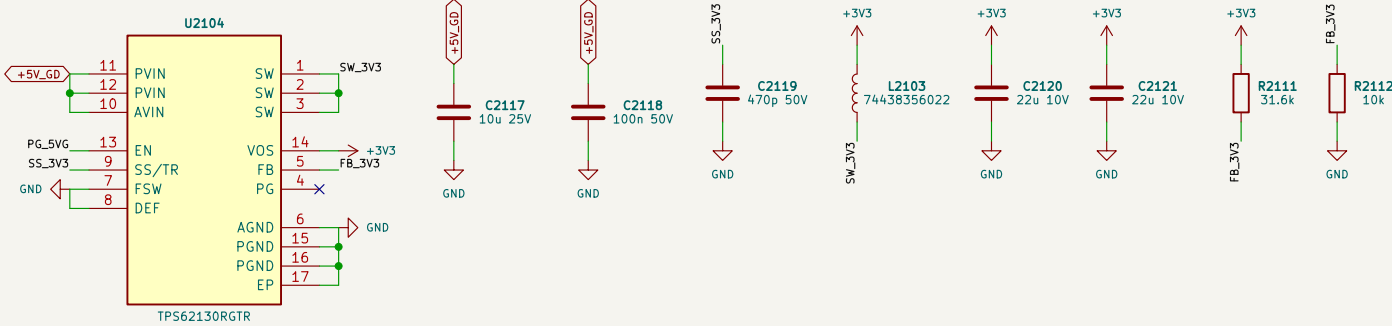
gate bias of phases 3–4, ctrlC2000 circuit (SNVSC40E table 9–1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93–5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



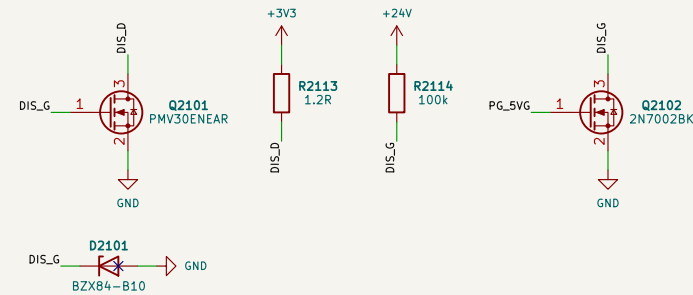
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3–4 is below 4.52–4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240–3.433 V (>= 3.20 V for the control board's reference), SS/TR 470pF COG; ramp-up 12–16 mV/us (F28003x VDDIO 8–100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90–94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO–DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX–T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 21.4–30.1 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12–36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 22

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic

Sheet: /19 Live 5 V / 3.3 V, SELV connector/

File: 19_supplies.kicad_sch

Title: PV-PWR-4 power board – Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026–10–05

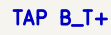
Rev: A2

KiCad E.D.A. 10.0.6

Id: 22/26

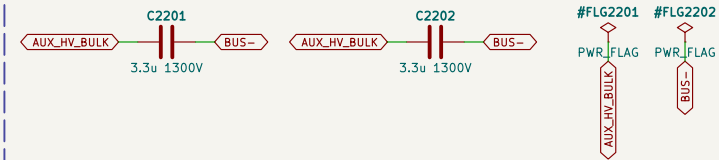


2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting



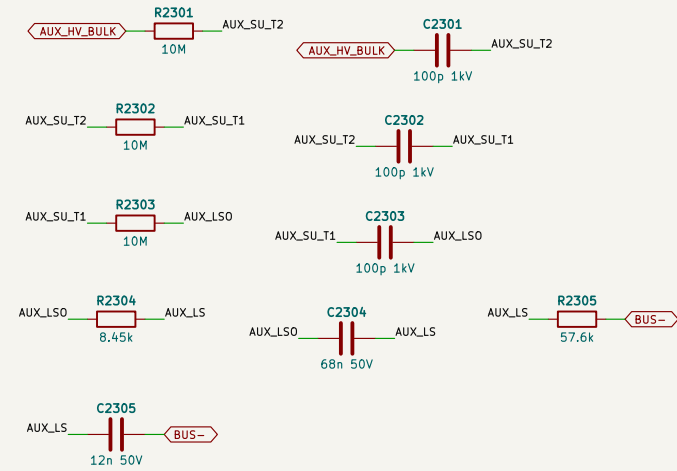
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.64 W each at the highest input current (0.48 A, 250 V)
Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



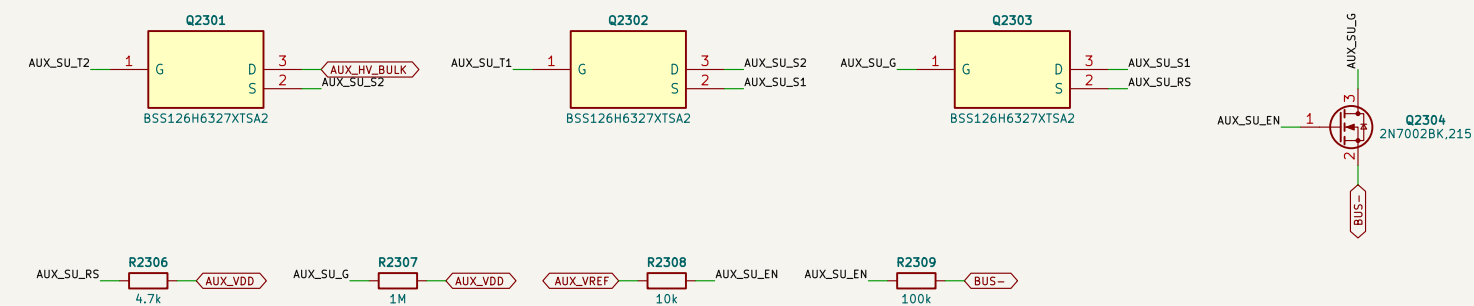
Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual → 1199 V (UR 1300 V); filter corner 546 Hz.

COMPENSATED TAP STRING



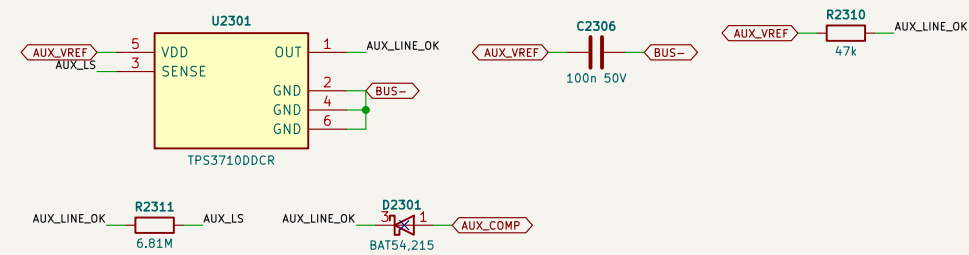
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



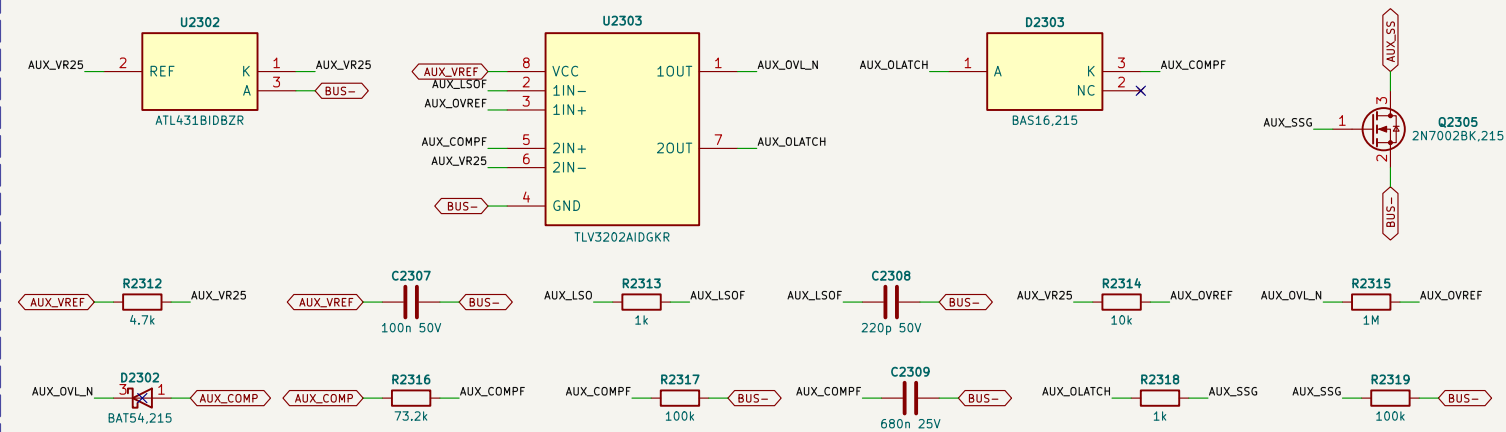
VREF high → start-up off (as AUX-HV).

LINE BROWN-IN



LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

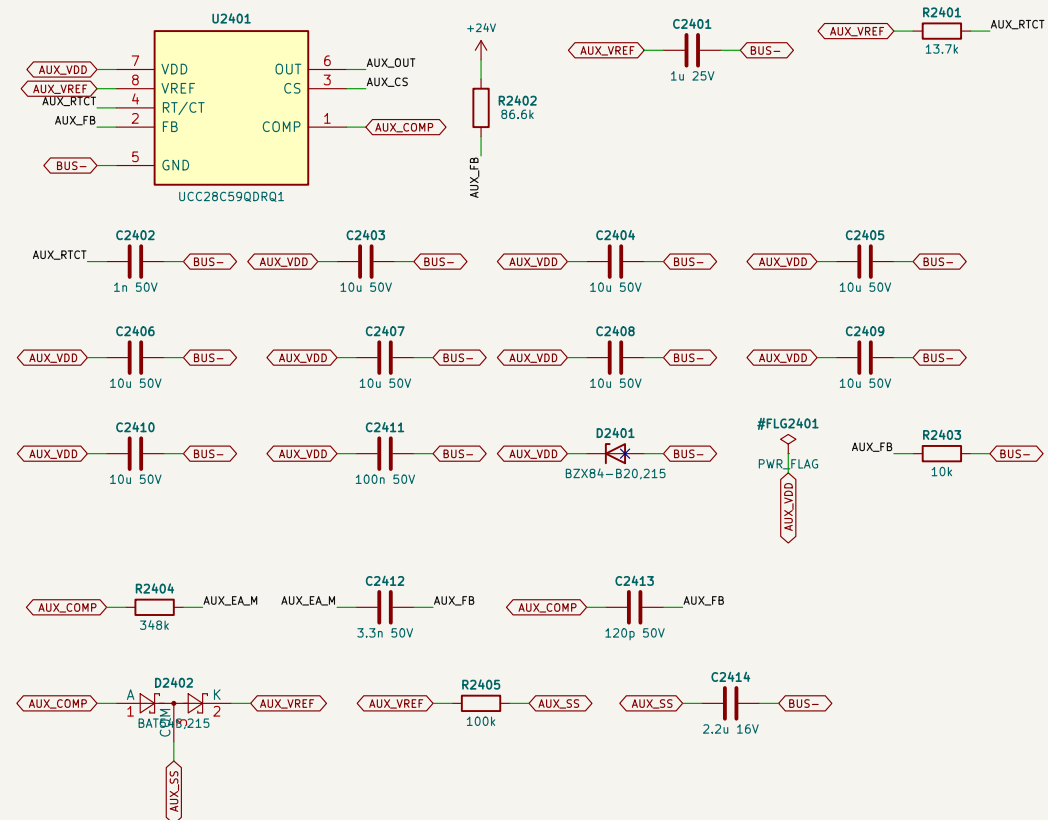
INPUT OV LOCKOUT AND OVERLOAD TIMER



Ch 1: OV lockout → COMP low, Ch 2: timer divider: trips above 113 / 175 / 278 W (no trip at the 112.2 W 1 s peak); timer 39–100 ms → latch → SS crowbar → hiccup.

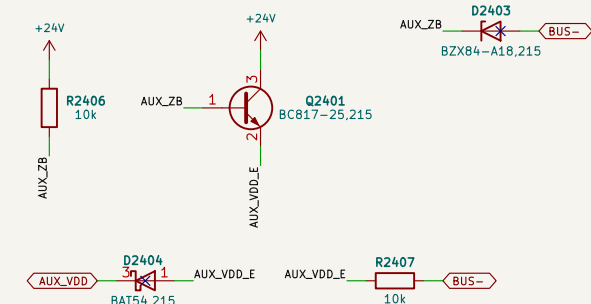
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 24 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic	
Sheet: /21 AUX75 start-up, brown-in, OV, timer/ File: 21_aux75_startup.kicad_sch	
Title: PV-PWR-4 power board – AUX75 start-up, brown-in, OV, timer	
Size: A3	Date: 2026-10-05
KiCad E.D.A. 10.0.6	Rev: A2 Id: 24/26

PWM CONTROLLER AND ERROR AMPLIFIER



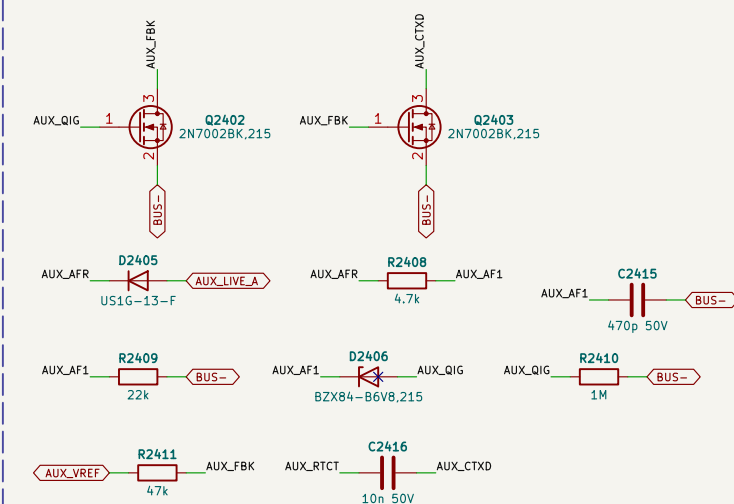
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 259–727 Hz, phase margin ≥ 60 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



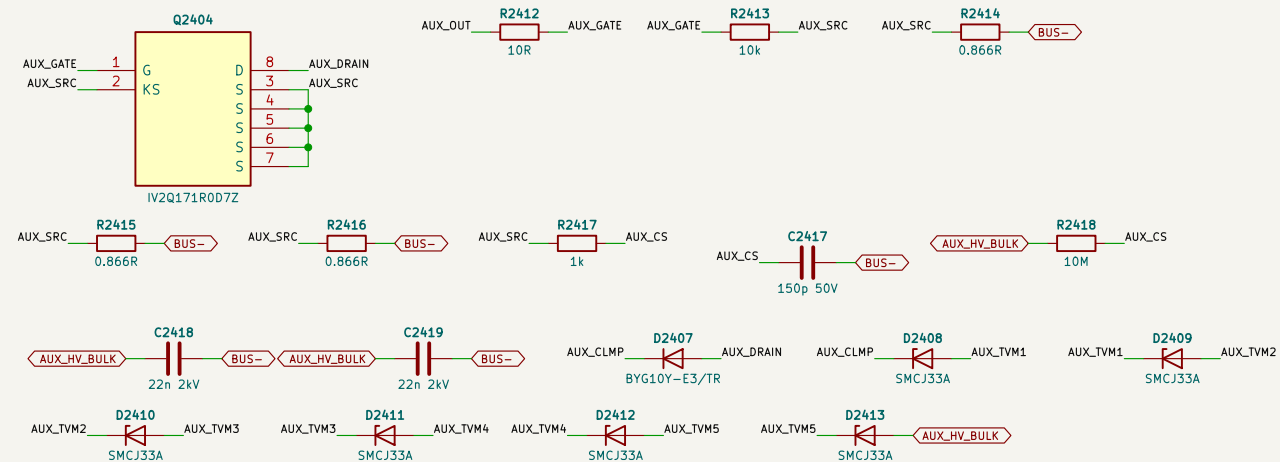
(30 W case, unchanged) follower Zener bias: $V_{DD} = \text{gate drive } 15.46\text{--}18.11 \text{ V}$ (BZX84-A18, $-40\text{--}85 \text{ }^{\circ}\text{C}$)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



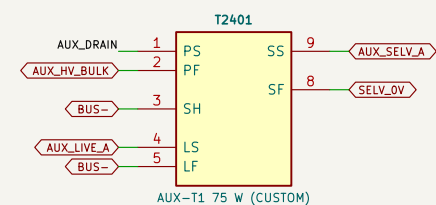
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



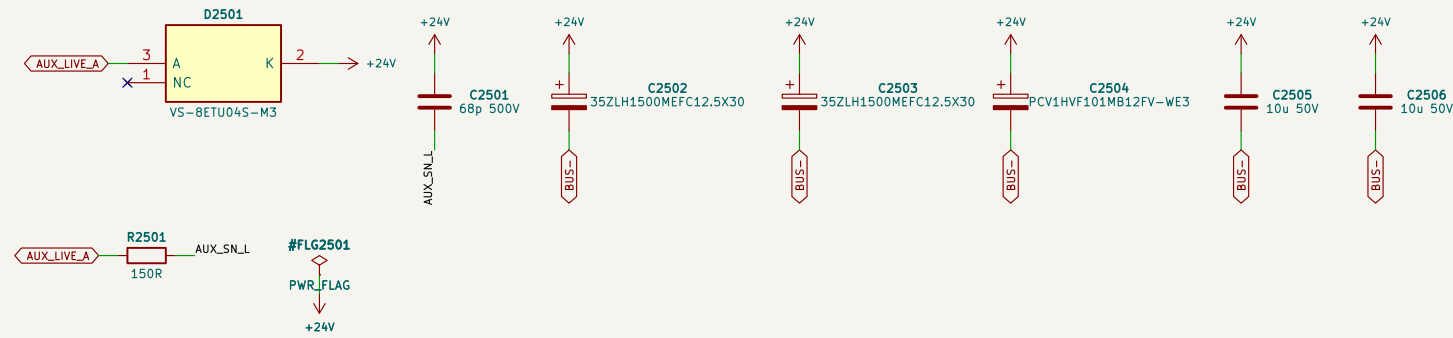
3 In parallel = 0.288667R (E96): lowest limit 3.04 A delivers 112.8 W (>= the 112.2 W 1 s peak) at every input; highest 4.07 A with the line feed-forward (4.46 A without) line feed-forward HV_BULK -> CS (CRH2512): CS offset 101 mV at 1000 V cancels most of the 155-250 nA CSJ44 overshoot, so the highest current limit no longer rises with the input Clamp: BYG10Y + 6 x SMCJ3A_VDS 1318 / 1422 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE’S REINFORCED SUPPLY BARRIER B1)



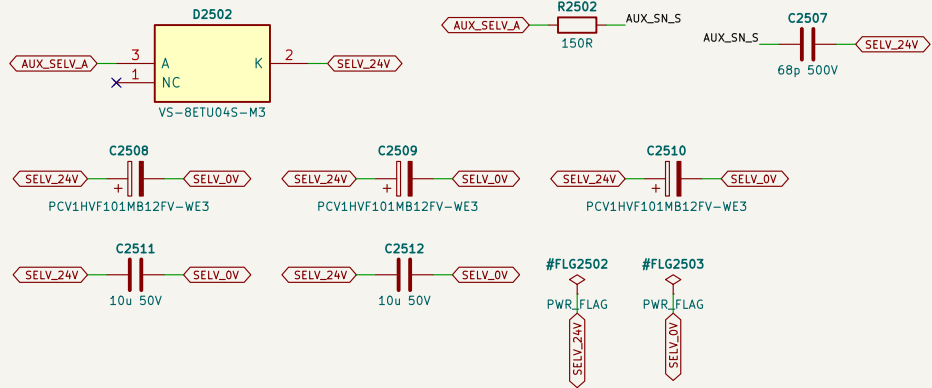
Lp 0.485 mH +/- 7 %: shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

LIVE 24 V RECTIFIER, HOLD-UP



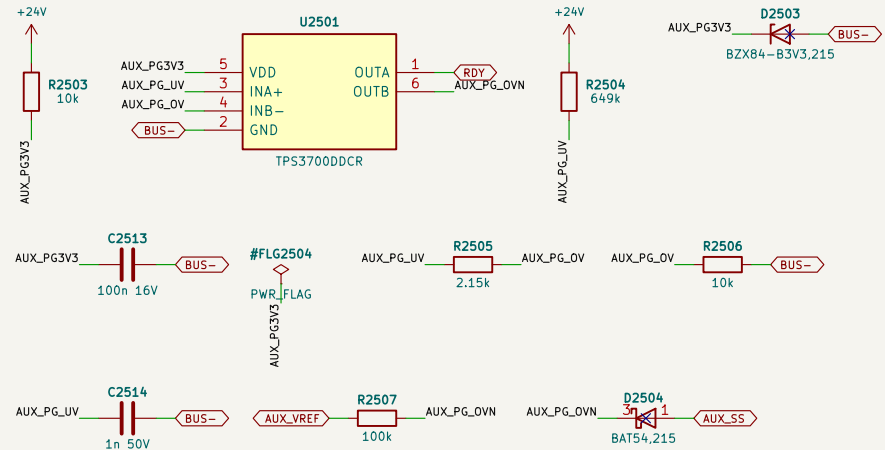
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02-21.81 V (status), OV rises 26.13-26.76 V (SS pulled low)